

B. PRODUCTION OF DEFECT CENTERS BY RADIATION

THE SELF-TRAPPED EXCITON

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Abstract—In cases of strong coupling of electrons or holes to the crystal lattice, a carrier may be self-trapped as a small polaron in its own lattice distortion field. A bound electron-hole pair involving such a carrier is generally described as a self-trapped exciton, and it may dramatically influence luminescence, energy transport, and lattice defect formation in the crystal. The phenomenon of exciton self-trapping is particularly common in metal halide and rare-gas crystals, where the strong exciton-lattice coupling can usually be ascribed to the possibility of covalent bond formation in the excited state of a crystal which does not admit such bonding in its ground state. Alkali halides are the most thoroughly studied of these materials. This paper will review experimental data, primarily spectroscopic, and theoretical treatments, primarily by methods developed for defects in solids, on the nature and consequences of self-trapped excitons.

Keywords: Self-trapped exciton, alkali halide, alkaline-earth halide, silver halide, rare-gas solid, quartz, luminescence, small polaron.

1. INTRODUCTION

In crystals where valence electronic excitation leads to a strong local deformation of the lattice, an exciton may be immobilized by its own lattice distortion field, i.e. self-trapped. Self-trapped carriers, both electrons and holes, are the limiting case of small polarons, in which the associated lattice distortion is primarily confined about a single lattice site [1–3]. Two qualitatively different polarons have been distinguished by Mott and Stincham [4]: “dielectric polarons” which result from the Coulomb interaction of a charge and a deformable polar lattice [1, 2], and “molecular polarons” for which the essential interaction is short-range chemical bond formation [4, 5]. In general, a real system may exhibit some of both the characters. See, for example, the discussion due to Gilbert on the separate contributions of lattice polarization, bond formation, and carrier localization to the energy of self-trapping in alkali halides [6, 7].

In this paper, we address primarily the “molecular” case of self-trapping, and concentrate on spectroscopic and photochemical properties of excitons which are self-trapped, rather than on transport properties of either charge carrier individually. The latter have been discussed in Refs 1–3. Self-trapping of excitons is the rule more than the exception in ionic halide crystals and in solid rare-gas crystals, but has been reported only rarely (e.g. SiO_2) [8] in other crystals. The ionic metal halides and rare-gas crystals share a similarity of valence structure in the ground state. They are composed of closed-shell rare-gas-like ions in the former case or rare-gas atoms in the latter, with the result that covalent bonding is minimal. The constituents of the crystals are held apart at ionic or atomic radii by the repulsive interaction of neighbors with filled valence shells. Valence excitation relieves

the closed shell condition, opening up the possibility of covalent bonding, most commonly as dimerization of near neighbors, though not always. Chemical bonding in the excited state provides a strong electron-lattice coupling mechanism which can trap the electronic excitation that made it possible. The self-trapped state may not form immediately since the valence excitation initially involves extended band states. However, in the unstable excited lattice described above, a thermal fluctuation of appropriate magnitude (essentially the self-trapping barrier discussed by Toyozawa [5]) will lead quickly to localization of the exciton in a self-trapped state of molecular dimensions that usually involves a covalent bond formed in a non-covalent solid.

The self-trapping phenomenon we have described above in terms of chemical bonding applies as well to self-trapping of holes as to excitons. The hole component of the exciton accounts primarily for the localization of self-trapped excitons in most of the systems which have been experimentally studied. This does not mean, however, that the electron has no part in determining the lattice distortion around the exciton. Its effect can be very dramatic, as we shall see.

In alkali halides [7, 9, 10], alkaline-earth halides, [11, 12], ammonium [13, 14] or phosphonium [15] halides, perovskite-structure halides of the KMgF_3 family [16, 17], and in rare-gas solids [18], the self-trapped hole has the form of a dimer X_2^+ (X_2^+ in rare gases), where X is a halide ion (rare-gas atom). In essentially all of the above cases, the valence band has pure halide (rare-gas) *p* character. The excited dimer configuration of the self-trapped hole is therefore quite reasonable, following the description given above. The valence band of silver halides has both halogen *p* and silver *d* character. In that case the

self-trapped hole localizes on a tetragonally distorted molecular complex comprising a central silver atom and six halide ions [19–21], a structure also consistent with the general bond-formation description given above.

Exciton self-trapping dramatically affects the intrinsic luminescence, energy transport properties, and radiation damage processes of the crystal in which it occurs. It is perhaps not surprising that crystals in which the electron–lattice coupling is so strong as to lead to self-trapping are also often susceptible to photochemical generation of lattice defects. Self-trapped excitons are considered the precursors of lattice defects formed by ionizing radiation in alkali halides, and in some ways qualify as transient lattice defects themselves. Self-trapped excitons and the lattice defects which evolve from them have also been implicated in laser-induced breakdown damage of alkali halides [22].

2. SELF-TRAPPED HOLES

In the Introduction, we saw that the majority of instances of self-trapping involve the strong electron–lattice coupling associated with covalent bond formation between lattice constituents that are reactive only in the excited state. The initiating event in such cases is most often the creation of a valence hole, and it is the hole which localizes on the new chemical bond, occupying the highest antibonding orbital of the bonded pair. Thus, formation of the self-trapped hole is in most cases fundamental to the existence of the self-trapped exciton, which in turn arises when a conduction electron is bound to the self-trapping site. We shall therefore begin with a survey of self-trapped holes in various materials. The most thoroughly studied cases are in the halide family. Hole transport described by hopping of small polarons has been reported in SiO_2 [23] and in MnO [24].

As long as the electrons are trapped elsewhere in a halide crystal, self-trapped holes are stable indefinitely at temperatures too low for them to hop from site to site. In the color center terminology, self-trapped holes are called V_k centers after the pioneering work of Känzig and Castner [9]. A review of V_k centers has been given by Kabler [7].

In Fig. 1, we summarize pictorially the lattice configurations of self-trapped holes in several materials in which they have been identified.

In all cases shown except silver chloride, the self-trapped hole is localized on a halogen dimer, X_2^- . This is isoelectronic with the rare-gas dimer, R_2^+ , on which the self-trapped hole is localized in solid rare gases exhibiting the “two-center” self-trapped exciton configuration [25, 26]. The states of a free X_2^- molecule ion are a useful model for labeling the states of a self-trapped hole in an alkali halide. The appropriate molecular orbital (MO) energy levels are illustrated schematically in Fig. 2, proceeding from separated atoms on the left, to single particle MOs

in the $D_{\infty h}$ symmetry of the diatomic molecule with neglect of spin–orbit interaction, then D_{2h} symmetry of the crystal field in NaCl structures and on the right side, MOs in $D_{\infty h}$ with spin–orbit interaction labeled by the total angular momentum parallel to the internuclear axis. The double group representation including both crystal field and spin is not shown. There is no net bonding interaction if all the orbitals are filled, but removal of an electron from the antibonding σ_u orbital leads to a net attraction even in the free X_2^- molecule ion. The molecular orbitals for the self-trapped hole in silver chloride have been discussed in Refs 19 and 20.

Allowed optical absorption transitions of the self-trapped hole are indicated by the arrows in Fig. 2. The $\sigma_u \rightarrow \sigma_g$ hole transition is by far the most prominent, accounting for the characteristic broad UV absorption band of V_k centers. In some crystals of the heavier halides, weaker infrared absorption due to $\sigma_u \rightarrow \pi_g$ hole transitions can be seen. Optical absorption spectra of self-trapped holes in a sampling of halide crystals are compared in Fig. 3.

A theoretical treatment of the self-trapped hole in KCl was developed by Jette *et al.* [28] who studied the structure of V_k centers in alkali fluorides and chlorides, incorporating a molecular orbital calculation of F_2^- and Cl_2^- embedded in a lattice. Pair potential and polarization effects were included and the total energy was minimized as a function of the X_2^- internuclear separation and of the positions of the six nearest-neighbor alkali ions in the (100) plane containing the X_2^- . They found little change in the structure of the molecule ion from the free state. This led to the concept of “molecule in a lattice” to describe the self-trapped hole. Tasker and Stoneham compared transition energies, line-widths, hyperfine constants, and g factors of self-trapped holes in pure and mixed halide crystals to values calculated within a molecular model, finding generally satisfactory agreement [29].

Although self-trapped holes are immobilized with respect to transport in band states, they can move by site-to-site hopping under thermal activation. The entity that hops is the hole and its accompanying bond between ions, i.e. bond switching rather than actual ion displacement. Polarized light has been very useful for studying the thermally-activated motion by revealing changes in bond orientation [7, 30, 31]. In the NaCl structure, movement of the hole to an adjacent self-trapping site can result in reorientation of the bond axis by 60 or 90°, in principle, and in translation with no reorientation (0°). Experiment has shown that 90° jumps are negligible in the NaCl-structure crystals studied. In CsCl-structure crystals, where 60° jumps are not possible, motion by 90 and 0° jumps is found. In CaF_2 , only 0° jumps are observed, but in SrF_2 and BaF_2 the transport of self-trapped holes proceeds by both 0° and 90° jumps.

In the theories of Toyozawa [5] and of Mott and Stoneham [4], there is an energy barrier between the

free exciton band states and the self-trapped state. It is generally presumed that such a barrier can also inhibit hole self-trapping, in principle. It may be viewed as the energy that must be invested in thermal fluctuations to begin hole localization and consequent bond formation. For self-trapped excitons, effects of this barrier are observed in alkali iodides at low temperature through its influence on branching between free and self-trapped luminescent states [32, 33] or between self-trapped exciton and impurity luminescence [34–36]. For holes, the barrier has been studied experimentally in silver chloride, where a barrier of 20 meV inhibiting hole self-trapping has been found [37]. In the experiments, doped impurity centers provided extrinsic trapping sites as alternatives to self-trapping. The relative yields of impurity-trapped and self-trapped holes vs temperature at temperatures still too low for hopping of self-trapped holes was monitored by EPR to determine the range of hole diffusion before self-trapping.

Even though no thermal activation barrier has yet been observed for hole self-trapping in alkali halides, free holes have a finite, and sometimes substantial, range before self-trapping. Hot-hole transport in alkali halides is seen in studies of scintillator response. The cross-section for electron-hole recombination at the activator generally exceeds the geometric cross-section of the activator by a substantial amount even at temperatures where hopping transport of self-trapped holes is frozen out. This is usually interpreted as indicating transport of hot holes before self-trapping.

Kadchenko and Elango have made a detailed analysis of silver-activated luminescence excited by X-rays in NaCl:Ag [38]. They deduced the following values from data at a temperature of 80 K: hot-hole range of about 720 Å before self-trapping, mean free path of 180 Å between scattering events, and average hot-hole velocity of 3×10^7 cm s⁻¹, suggesting a mean time for self-trapping of 2.3 ps [39].

The time for hole self-trapping has not been directly resolved yet, but picosecond optical measurements have established upper limits. The formation of V_k centers in KI at both room temperature and liquid helium temperature appears instantaneous within the limits of deconvoluting 40-ps pulses [40]. If a barrier against hole self-trapping exists in KI, it is small indeed. In NaCl at 10 K, the rise time of self-trapped exciton absorption (implying prior hole self-trapping) also appeared instantaneous within the limit of deconvoluting 30-ps pulses [41]. A rough estimate sometimes given for the expected time for self-trapping of holes in alkali halides is 10^{-13} s (100 fs). The longitudinal optical phonon period in NaCl is 125 fs. This may be a reasonable estimate for self-trapping of holes at the top of the valence band in a crystal near room temperature. The range deduced by Kadchenko and Elango [38] suggests a longer time for cooling of hot holes. Direct optical measurements in the alkali halides with subpico-

second time resolution are now possible [42, 43], and results on the time for hole self-trapping should be interesting.

3. SELF-TRAPPED ELECTRONS

In the earliest and most general treatments of carrier self-trapping [1, 2], the electron has usually been the carrier considered. Yet in nature, the occurrence of self-trapped electrons seems to be considerably rarer than the occurrence of self-trapped holes, at least when the prevalence of self-trapped holes among the metal halides is counted. Self-trapped electrons (i.e. small electron polarons) have been identified from transport properties in orthorhombic sulfur [44], As₄S₄ [45], and in some rare gases [18], for example.

Gilbert has discussed the energetics of self-trapping, showing in general terms why electrons are less likely than holes to self-trap in at least the halides [6, 7]. Three contributions to the self-trapping energy were considered. When the carrier localizes, the surrounding crystal polarizes around the charge, giving up a polarization energy, E_{pol} . In the molecular case such as for self-trapped holes in alkali halides, there is also formation of a chemical bond, involving an additional negative energy change, E_{bond} . Once localized in real space, the carrier is delocalized in k -space with an energy near the middle of the band. Since the free carrier in equilibrium in an insulator or semiconductor would have occupied a state near the band edge before localization, the positive localization energy, E_{loc} , is about half the width of the energy band involved.

The energy E_{st} for self-trapping of holes in KCl, for example, is

$$E_{\text{st}} = E_{\text{pol}} + E_{\text{loc}} + E_{\text{bond}} \\ \approx -0.5 \text{ eV} + 0.5 \text{ eV} - 1.0 \text{ eV} = -1 \text{ eV}, \quad (1)$$

in rough agreement with other estimates of the self-trapping energy. On the other hand, the width of the conduction band in alkali halides is substantially larger than the valence band width, leading to $E_{\text{loc}} \approx +2$ eV for electrons in KCl. The polarization energy is about the same for electrons and holes, whereas the bond energy for hypothetical K_2^+ formation (the self-trapped electron analog of Cl_2^-) was estimated to be only about -0.3 eV [6, 7]. Thus the net energy for electron self-trapping in KCl is positive. It is the relatively narrow valence band and the unusually strong bonding of the halogen₂⁻ molecular ion which accounts generally for the prevalence of hole self-trapping in metal halides.

A totally different kind of self-trapping of electrons was discovered in rare-gas solids, however. It is found through studies of photoemission that conduction electrons in solid Ne and marginally in solid Ar have negative electron affinity [46]. This means that an electron which is excited to the bottom of the conduc-

tion band can be photoemitted with a finite kinetic energy (about 1 eV in solid Ne). This repulsion of a conduction electron by the lattice can be understood in conjunction with the formation of the "electron bubble" observed in liquid Ne among others.

This formation of the "electron bubble", detected through mobility studies, is interpreted in terms of the short-range interaction between the excited electron and the surrounding atoms. It is shown that the overall interaction is repulsive in these lighter atom condensed rare gases [47]. This repulsive interaction leads to the formation of a small cavity around the electron which in turn immobilizes the charge. The cavity contains in general several atomic volumes.

4. OBSERVED PROPERTIES OF SELF-TRAPPED EXCITONS

(A) Luminescence

Recombination of electrons with self-trapped holes in most metal halides at low temperature yields one or more broad luminescence bands which are Stokes-shifted by several electron volts from the band edge. Kabler [48] and Murray and Keller [49] first identified the luminescence and established the existence of self-trapped excitons in alkali halides. Key data in-

cluded the polarization of the transitions referred to the σ bond axis of preferentially-aligned self-trapped holes. In all alkali halides, there is at least one π -polarized luminescence band which has a relatively long lifetime and large Stokes shift. Kabler and Patterson identified the π -polarized emission as spin-forbidden recombination from a triplet state of the self-trapped exciton, made partially allowed (and π -polarized) by spin-orbit induced admixture of singlet Π character in the excited state [50]. In some of the alkali halides, there is a σ -polarized, short-lived luminescence band with a Stokes shift 1 or 2 eV smaller than for the π luminescence. This is attributed to recombination from a singlet state of the self-trapped exciton. However, the 1 or 2 eV difference in peak energies of the σ and π luminescence is much larger than the exchange splitting between singlet and triplet states of the self-trapped exciton, e.g. 7.6 meV in KBr [51, 52]. In more than half of the alkali halides, there is no σ luminescence at all, even though the spin-forbidden π luminescence is quite strong. Satisfactory explanations for the origin of the σ luminescence and its relation to the better-understood π luminescence states are still lacking. It has generally been thought to originate from a higher electron orbital of a_{1g} symmetry, usually denoted a_{1g}^* (see note added in proof).

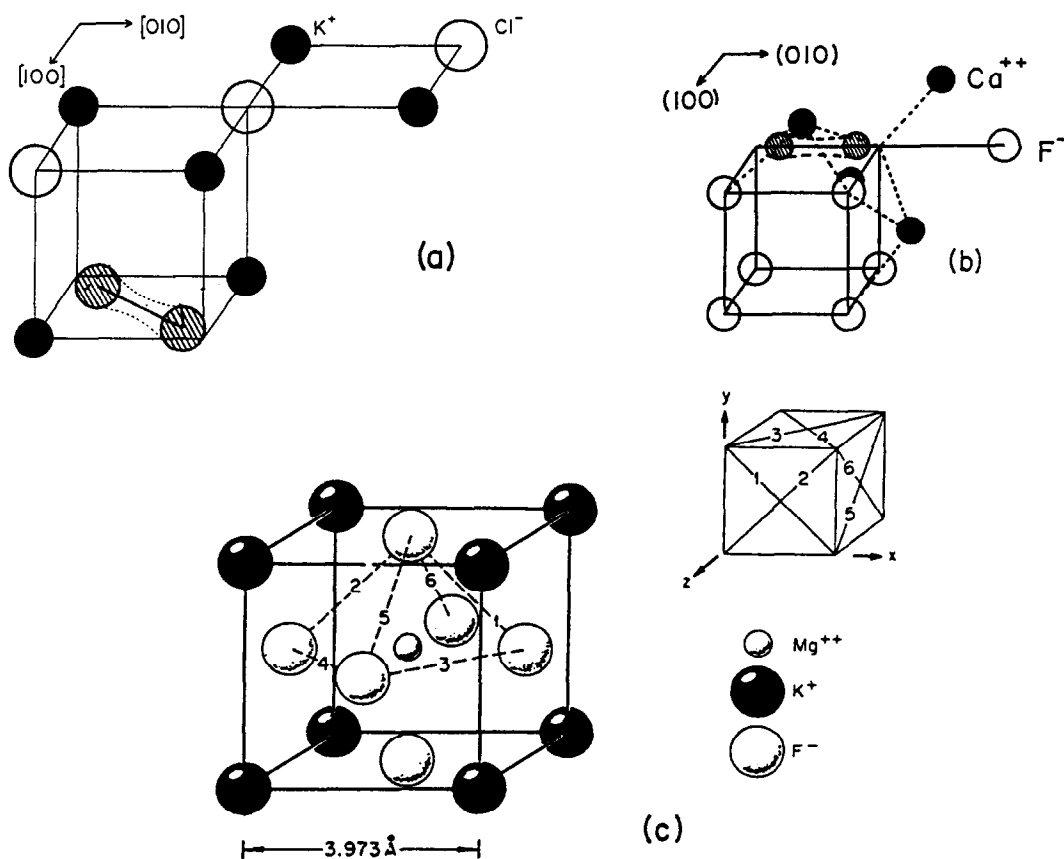


Fig. 1(a)-(c)

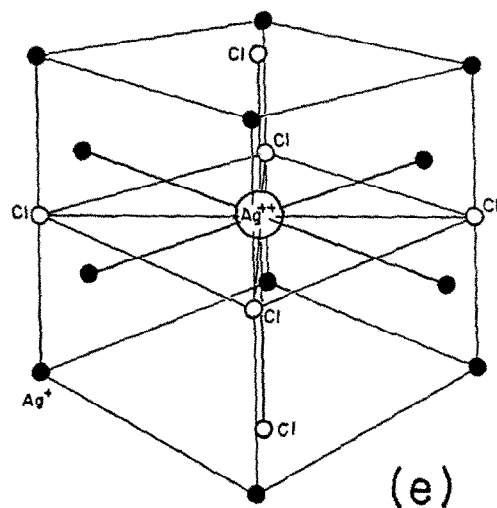
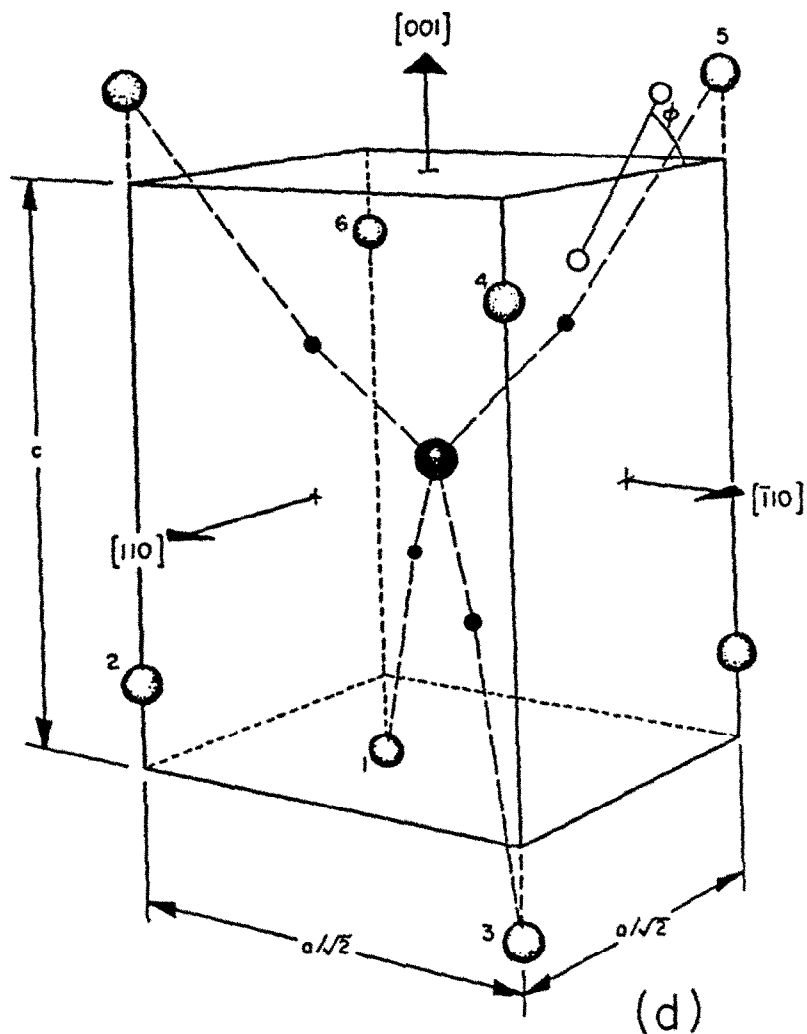


Fig. 1(d)-(e)

Fig. 1. Lattice configurations of self-trapped holes (V_k centers) are illustrated for (a) KCl, (b) CaF_2 , (c) KMgF_3 (after Ref. 16), (d) PH_4I (after Ref. 15), and (e) AgCl (after Ref. 19). In KCl and CaF_2 , the bonded halogen pair is shown shaded. In PH_4I , the small circles at the upper right show the positions to which ions 4 and 5 relax to form the I_2^- center. In KMgF_3 , the hole can self-trap on any one of the fluoride ion pairs labeled 1-6, followed by inward relaxation of the pair analogous to that shown in CaF_2 . In AgCl , the self-trapped hole localizes primarily on the central silver ion, becoming Ag^{2+} .

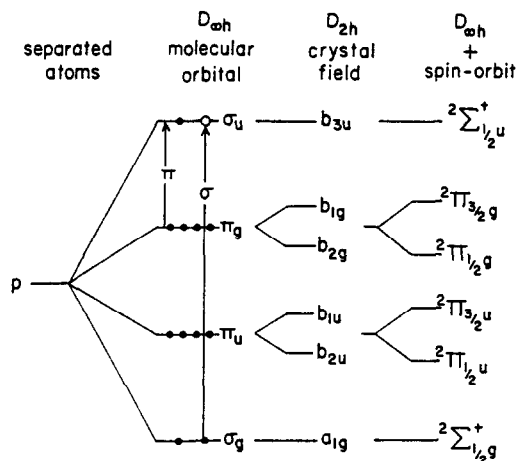


Fig. 2. Molecular orbital levels appropriate to the X_2^- configuration of the self-trapped hole. Proceeding from the left, we show the isolated atomic halogen p level, molecular orbital levels for the diatomic species in $D_{\infty h}$ symmetry with neglect of spin-orbit interaction, orbitals in D_{2h} symmetry appropriate to the NaCl-structure crystals, and states of total angular momentum parallel to the molecular axis assuming $D_{\infty h}$ symmetry.

Energy levels of the self-trapped exciton are illustrated in Fig. 4 in a format roughly analogous to that for self-trapped holes in Fig. 2. Luminescence occurs upon return of the excited electron to the topmost

hole orbital labeled σ_u or b_{3u} in $D_{\infty h}$ and D_{2h} symmetry, respectively. The halogen spin-orbit interaction mixes some π_u (i.e. b_{1u} , b_{2u}) orbital character into the σ_u hole orbital, accounting for the π polarization and partially-allowed nature of recombination with an electron occupying the excited orbital denoted $1s\sigma_g(a_{1g})$ in Fig. 4, in a spin triplet state with the unpaired electron in the σ_u valence orbital. State designations including the electron and hole orbitals (in parentheses) are shown in Fig. 4(b), for both $D_{\infty h}$ and D_{2h} symmetry, as well as C_{2v} to be discussed later. The above referenced hypothesis for origination of σ luminescence from an orbitally distinct $2s\sigma_g(a_{1g}^*)$ electron orbital rather than the singlet state associated with $1s\sigma_g(a_{1g})$ is represented in Fig. 4. Upward arrows represent optical transitions of the self-trapped exciton, to be discussed later. On the right side of Fig. 4, orbitals and states appropriate to C_{2v} symmetry are shown. These will be used later in discussions of off-center relaxation of the self-trapped exciton.

Figures 5–7 show luminescence spectra of self-trapped excitons in a selection of halide crystals and in a solid rare gas. Tables 1 and 2 summarize properties of self-trapped exciton luminescence in alkali halides and in other crystals.

Self-trapped excitons in alkali halides doped with heavier halogen impurities have been the subject of

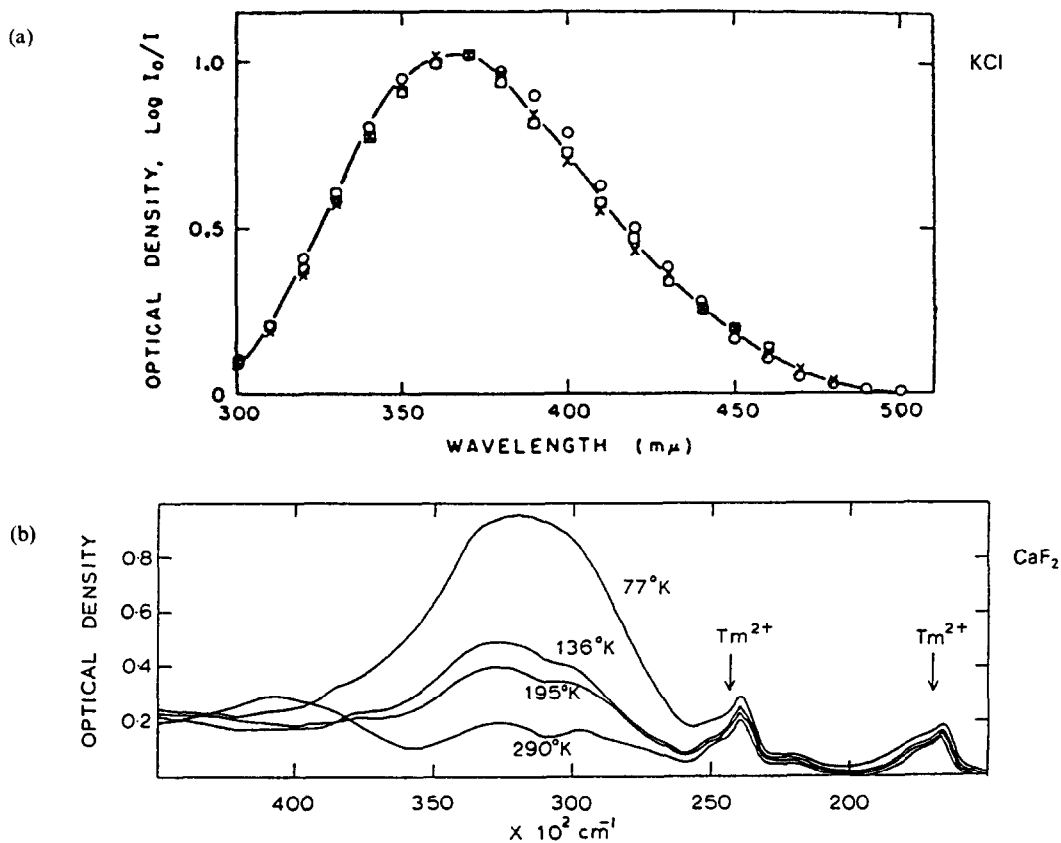


Fig. 3(a)–(b)

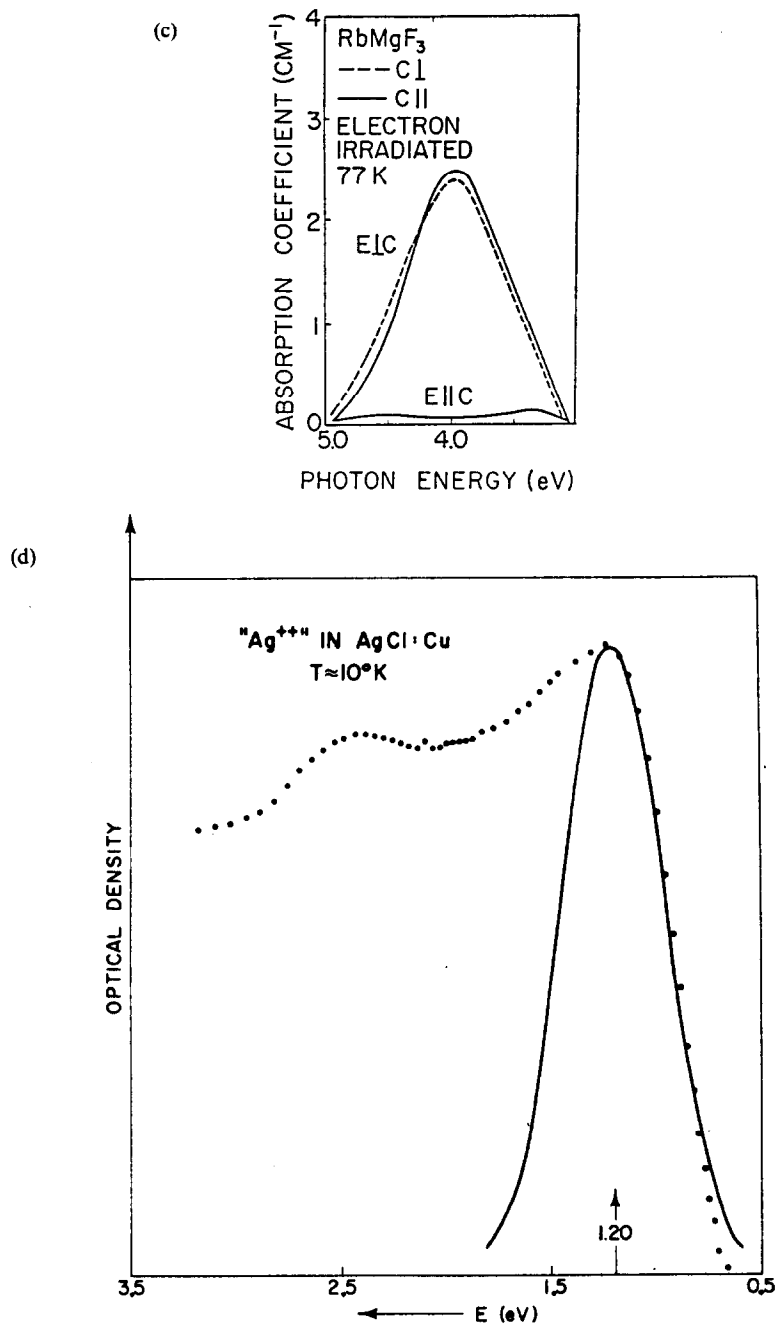


Fig. 3(c)-(d)

Fig. 3. Optical absorption spectra of self-trapped holes (V_k centers) in (a) KCl, (b) CaF_2 , (c) RbMgF_3 , and (d) AgCl. The spectra in (b) show the effects of annealing at the indicated temperatures after X-irradiation at 77 K of Tm-doped alkaline earth fluorides. In AgCl, shown in (d), the solid curve indicates the 1.2-eV transition of the self-trapped hole. There is also a 1.8-eV transition of the self-trapped hole. Much of the high energy absorption shown by the experimental points is associated with the Cu dopant. The data are from (a) Delbecq *et al.* [27], (b) Beaumont *et al.* [12], (c) Koumvakalis and Sibley, [17] and (d) Marquardt *et al.* [21].

interesting investigations [71]. There are two particularly notable aspects of the halogen impurity luminescence. One is the efficient creation of the host lattice F-H pair after exciting the impurity ion with photons of energy smaller than the host band gap. The other is the triplet state recombination luminescence of

the impurity-associated STE, such as ICl^- in KCl:I. Kan'no and co-workers have extended the latter studies over a wide range of combinations of $\text{MX}:\text{Y}$, where M is the alkali ion and X, Y are the halide ions [72]. Among other things they found that impurity-dimer associated self-trapped excitons such as

($I_2^- + e$) in MCl:I are characterized by small Stokes shifts and short lifetimes just like the self-trapped excitons in pure NaBr and NaI. We will discuss this in the section on theoretical interpretations.

Figure 8 shows the luminescence spectrum ascribed to self-trapped excitons in SiO_2 [67, 68, 70, 73]. A model proposed for the self-trapped exciton in SiO_2 is represented in Fig. 9, comprising essentially a close vacancy-interstitial pair on the oxygen sublattice, where the interstitial is bonded in a neutral oxygen molecule and two electrons are trapped on the vacancy [67, 68].

(B) Quenching of luminescence

Table 1 lists parameters for the thermal quenching of STE luminescence, represented by the rate of branching to a non-radiative de-excitation channel,

$$v = v_0 \exp[-E/kT]. \quad (2)$$

If this non-radiative channel competes with a constant radiative (first-order) decay rate τ_R^{-1} , the observed temperature-dependent lifetime of luminescence should be given by

$$\tau(T) = [\tau_R^{-1} + v_0 \exp(-E/kT)]^{-1} \quad (3)$$

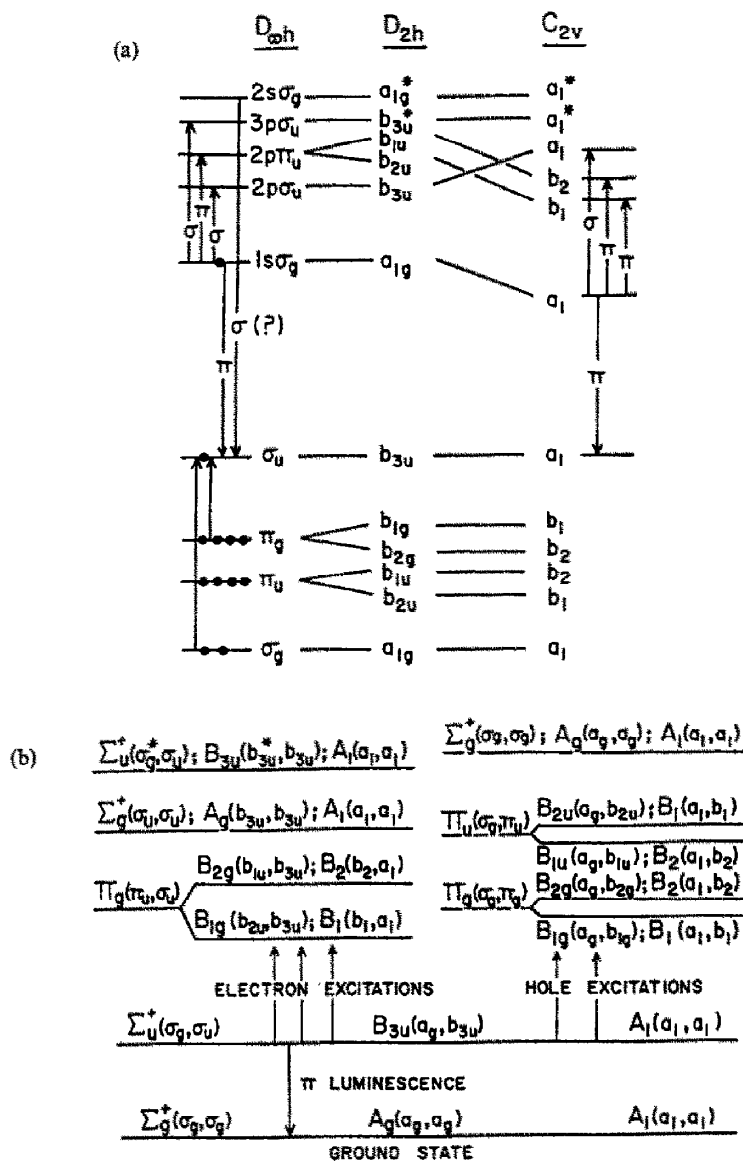


Fig. 4. In (a), electron and hole orbitals of the self-trapped exciton in alkali halides are shown, labeled by designations appropriate to $D_{\infty h}$, D_{2h} , and C_{2v} symmetries. Arrows indicate absorption and luminescence excitations, with the designated polarization referred to the X_2^- symmetry axis. In (b), each state is labeled from left to right in the three symmetries, $D_{\infty h}$, D_{2h} , and C_{2v} . Each state designation is followed in parentheses by the molecular orbitals of the electron and hole, respectively. That is, state (electron, hole) is designated in each case. Luminescence transitions (π) and hole and electron absorption transitions are shown by arrows. The diagram is only schematic with respect to transition energies.

and the steady-state luminescence intensity would have the temperature dependence

$$I(T) = I_0 \tau(T) / \tau_R. \quad (4)$$

The barrier E has been suggested to characterize

thermally-induced non-radiative transitions from the luminescent state directly to the ground state at the point of near crossing of the configuration coordinate curves of the two states [53, 74]. However, it has been shown more recently in Refs 74–78 that in alkali halide crystals which exhibit correlation of STE

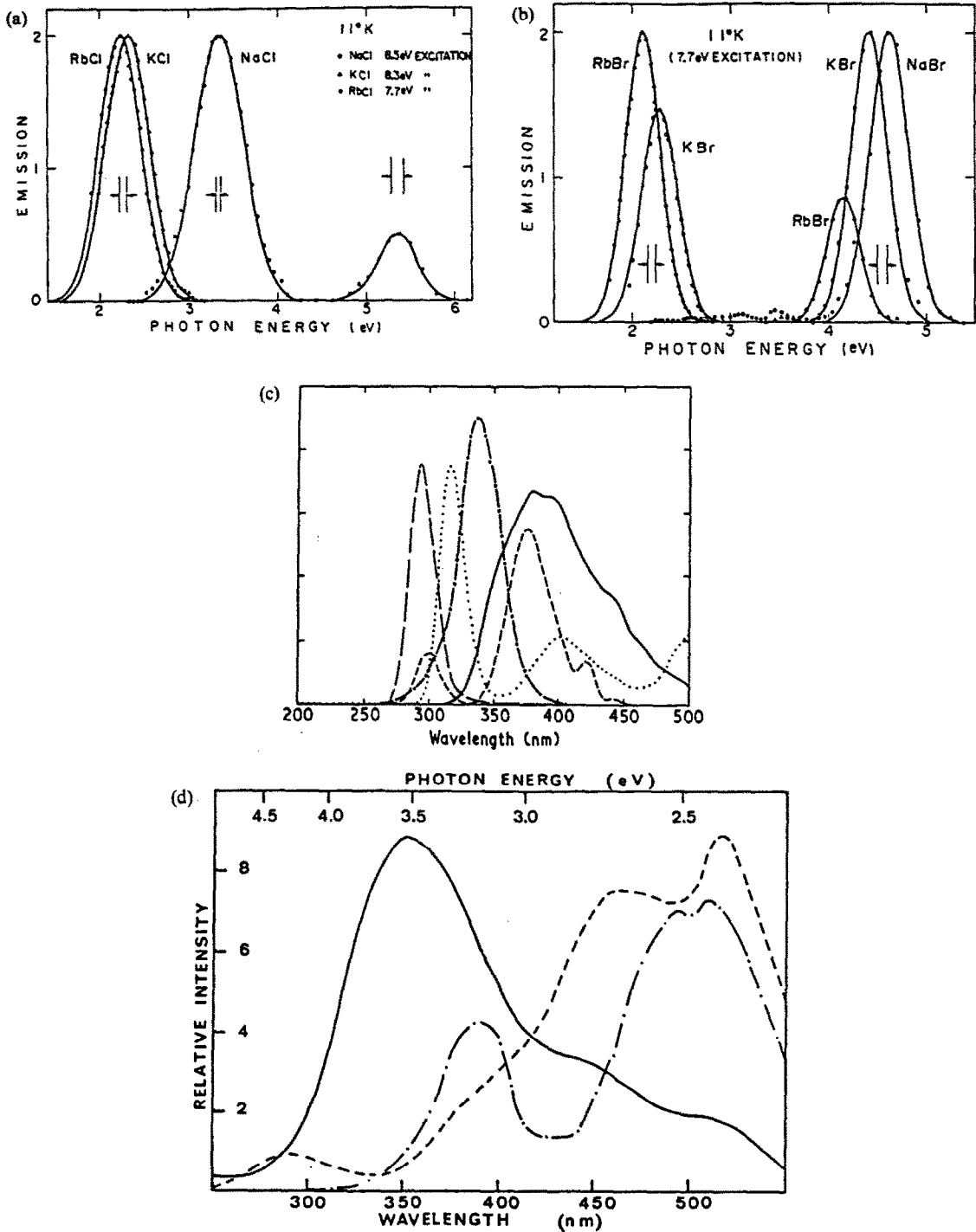


Fig. 5. Luminescence spectra of self-trapped excitons in alkali halides. The data in (a) and (b) are from Ref. 53, (c) from Ref. 54, and (d) from Ref. 55. In (c), LiI is represented by the full curve; NaI by the large broken curve; KI by the small broken curve; and RbI by the dotted curve. In (d), LiF is represented by the full curve; NaF by the broken curve; and KF by the chain curve.

π luminescence quenching and F-center formation yield, the activation energy in eqns (2) and (3) must describe the barrier between the STE luminescent state and the F-H pair on the same adiabatic potential surface. In other crystals, the activation barrier may still govern crossing to the ground state. The temperature dependence of STE luminescence intensity and lifetime is shown for a number of crystals in Figs 10 and 11, respectively.

(C) Yields

At temperatures well below the quenching temperature, the STE luminescence efficiency in alkali halides is in the range from about 0.1% to almost unity, depending on the material. Absolute yields are difficult to determine accurately, so that efficiencies relative to a standard are usually reported. NaI yields almost one photon of self-trapped exciton luminescence per electron-hole pair produced at 78 K [80]. Using NaI as a reference standard, Purdy and Murray determined a luminescence yield of about

0.1 photon of self-trapped exciton luminescence per electron-hole pair at 7 K in KCl [81], which is one of the less efficient emitters among the pure alkali halides. In general, the alkali halides with heavier halogens and lighter alkalis have the higher luminescence efficiencies. These are the same crystals having typically the greatest singlet admixture into the luminescent triplet state and therefore the shortest triplet lifetime. They generally have the higher quenching temperatures. Intrinsic luminescence yields in pure alkali halides referenced to CsI as the standard of unity efficiency have been measured by Pooley and Runciman [54], and are listed in Table 1.

(D) Ultraviolet excitation spectra

The STE luminescence yield has been measured as a function of UV and soft X-ray photon energy in alkali halides [53, 82–84] and in alkaline earth halides [85, 86]. Examples of XUV excitation spectra for several alkali halides [82] are shown in Fig. 12. Such excitation spectra are found to be dominated by

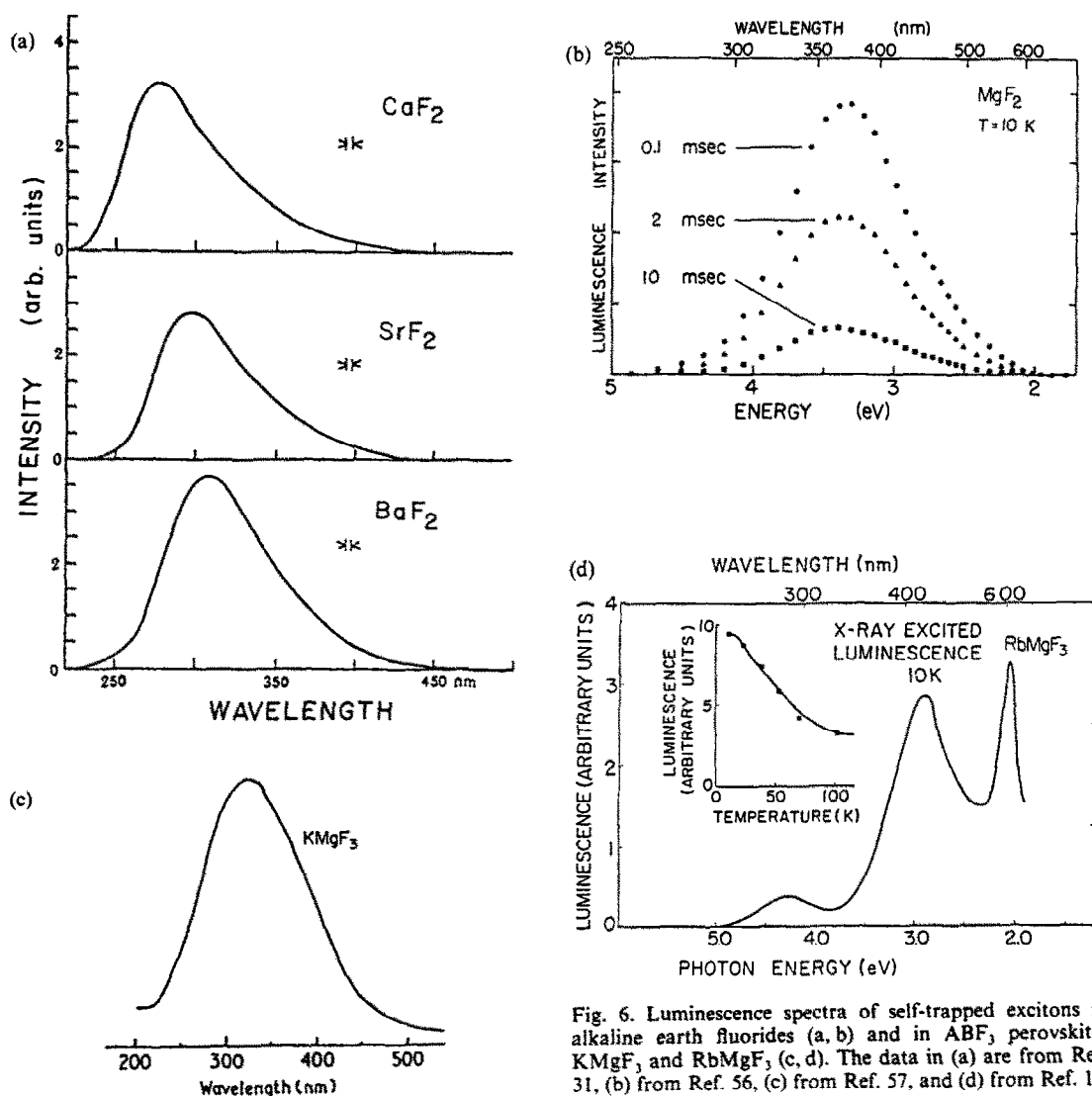


Fig. 6. Luminescence spectra of self-trapped excitons in alkaline earth fluorides (a, b) and in ABF_3 perovskites KMgF_3 and RbMgF_3 (c, d). The data in (a) are from Ref. 31, (b) from Ref. 56, (c) from Ref. 57, and (d) from Ref. 17.

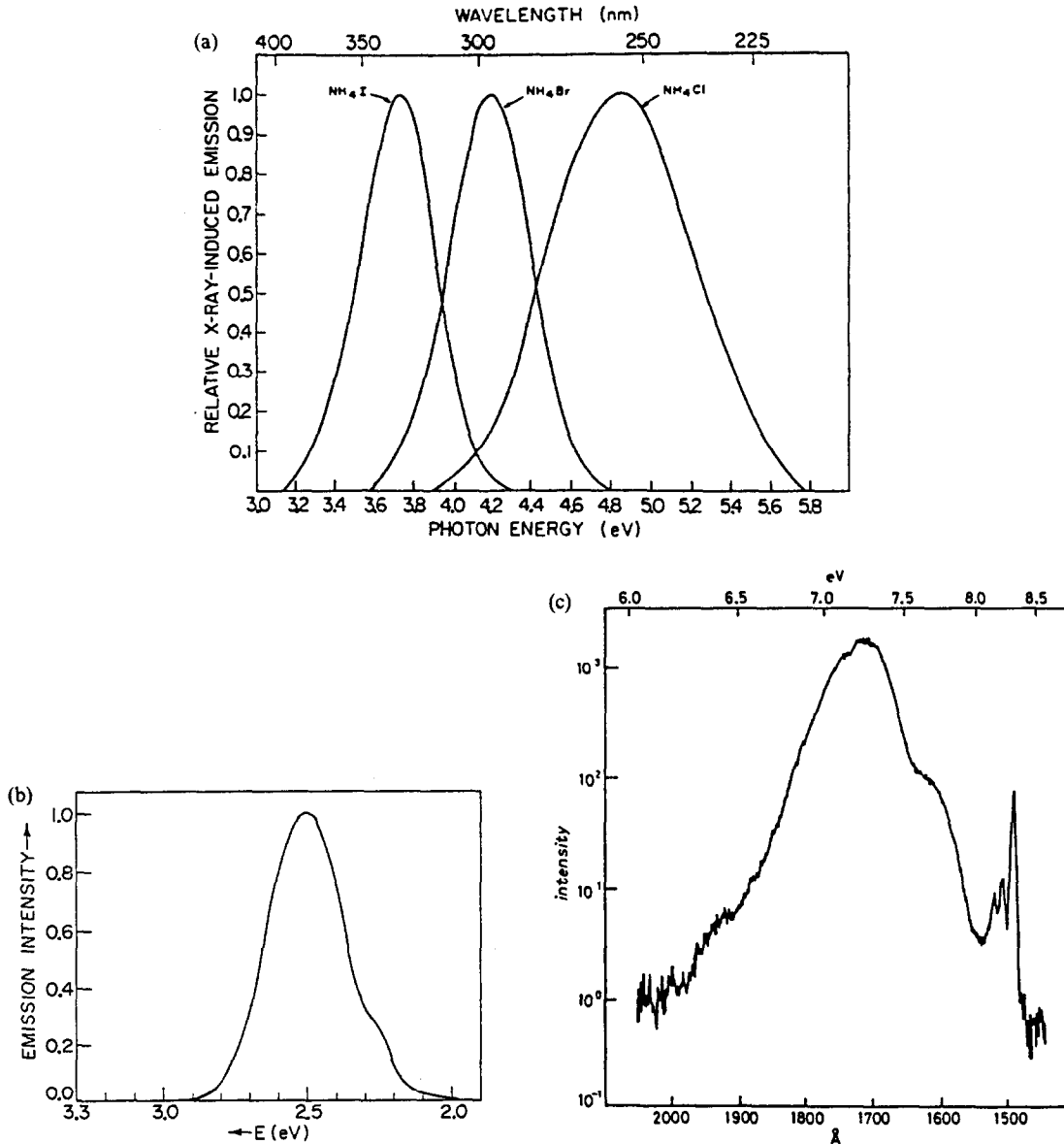


Fig. 7. Luminescence spectra of self-trapped excitons in ammonium halides (a), in silver chloride (b), and in solid xenon (c). The data in (a) are from Ref. 58, and (b) are from Ref. 19. The spectrum of electron-excited solid Xe at 60 K (c) exhibits free-exciton Xe luminescence and Kr impurity luminescence near 1500 nm, in addition to the two broad bands which are attributed to the molecular-type (Xe_2^+ dimer) self-trapped exciton [26].

anticorrelation between luminescence yield and strong absorption peaks, as well as a general stepwise increase of yield at multiples of the band gap. The anticorrelation behavior suggests a dead layer at the surface, where STE luminescence yield is presumed to be substantially diminished, as well as the effect of strong peaks in reflectance of the exciting light.

Excitation spectra of STE luminescence in the region of the lowest exciton peaks and band edge are shown in Fig. 13. Excitation of the triplet (π) luminescence begins on the low energy edge of the first exciton absorption peak and is strongest around that peak, except for a sharp dip in efficiency coincid-

ing with the peak itself [53]. Excitation of the singlet (σ) luminescence begins consistently near the band edge or onset of $2p$ exciton absorption, as identified from two-photon absorption spectra [87] as shown.

(E) Free exciton luminescence and the self-trapping barrier

Luminescence spectra of four alkali iodides including the region of the first exciton absorption peak are shown in Fig. 14, from the work of Nishimura *et al.* [33]. Nearly coincident with the lowest exciton absorption transition is a narrow luminescence peak attributed to zero-phonon optical

Table 1. Peak energy of the 1s exciton, luminescence band peak, polarization, full-width at half-maximum, radiative lifetime at nominal liquid helium temperature, quantum yield relative to CsI, temperature of quenching (to 10% of maximum), and thermal quenching parameters E_0 and ν_0 appropriate to eqn (3) are listed for self-trapped excitons in alkali halide crystals†

Alkali halide	1 s exciton peak (eV)†	STE emission peak (eV)	Pol.	Width (eV)	Lifetime at LHeT (s)	Quantum yield (CsI = 1)§	T_{quench} (10%) (K)††	E_0 (eV)††	ν_0 (s ⁻¹)§§
LiF	12.9	3.56§	π			5×10^{-4}			
		5.8§							
		3.5¶							
NaF	10.66	2.7¶	π		1.45×10^{-2}	2.4×10^{-4}			
		2.4¶							
		4.27							
KF	9.88	2.48¶	π	0.45	2.8×10^{-6}	2.5×10^{-3}			
		3.06§							
		3.2¶							
RbF	9.54	3.10§	π	0.36		2.6×10^{-3}			
		5.32§							
		3.13§							
CsF	9.27	5.96§	π	0.33		7.6×10^{-3}			
		3.13§							
		5.1 $\times 10^{-3}$							
LiCl	8.67	4.18§	π	0.43	1.2×10^{-4}	1.9×10^{-2}			
		3.47§							
		3.1 $\times 10^{-4}$							
NaCl	7.96	5.47§	π	0.68	3.1×10^{-4}	2.3×10^{-2}			
		5.61†							
		2.8×10^{-9}							
KCl	7.79	3.38††	σ	0.59	$\leq 5 \times 10^{-9}$	1.7×10^{-2}			
		3.35††							
		2.95×10^{-4}							
RbCl	7.54	5.35††	π	0.63			80	0.099	4.7×10^{10}
		3.38							
		3.2×10^{-4}							
CsCl	7.85	2.54§	π	0.53	3.4×10^{-4}	2.1×10^{-3}	70	0.032	3.2×10^{11}
		2.32††							
		5×10^{-3}							
LiBr	7.23	2.31††	π	0.57	8.9×10^{-3}	6.7×10^{-3}	85	0.1118	1.7×10^{11}
		2.41§							
		5.5×10^{-3}							
NaBr	6.71	2.27††	π	0.25	1.9×10^{-9}	7.0×10^{-4}	19	0.012	2×10^9
		2.23††							
		5.1×10^{-6}							
		7.52§	σ	0.54	1.9×10^{-9}	9.5×10^{-3}	22	0.009	1.8×10^5
		5.07§							
		4.6×10^{-7}							
		3.95§	π	0.45	4.9×10^{-7}		170	0.16	1.2×10^{12}
		5.30§							
		4.60 ††							
		4.65§	σ	0.44					
		4.60††							
		4.62 ††							

KBr	6.71	2.44§	π	0.41	1 × 10 ⁻⁴	1.4 × 10 ⁻²	43	0.037	1.5 × 10 ⁹
		2.27††	π		1.3 × 10 ⁻⁴				
		2.28‡‡	π	0.44					
		4.40§	σ	0.44	3.3 × 10 ⁻⁹	2.9 × 10 ⁻²			
		4.42††	σ		~9 × 10 ⁻⁹				
RbBr	6.64	4.42††	σ	0.44			55	0.023	1.1 × 10 ¹¹
		4.42‡‡	σ						
		2.36§	π	0.47	1.5 × 10 ⁻⁴	3.8 × 10 ⁻³			
		2.10††	π		1.8 × 10 ⁻⁴				
		2.11‡‡	π	0.42					
CsBr	6.83	4.13§	σ	0.39	3 × 10 ⁻⁹	3.4 × 10 ⁻²		0.008	—
		4.20††	σ		11 × 10 ⁻⁹				
		4.15‡‡	σ	0.40					
		3.55§							
				0.28	1.5 × 10 ⁻⁵	1.9 × 10 ⁻²			
LiI	5.94			0.40	3.3 × 10 ⁻⁹	8.1 × 10 ⁻³			
		3.27§	π		5.0 × 10 ⁻⁶	6.4 × 10 ⁻³			
		4.24§	π	0.38	5.7 × 10 ⁻⁷	2.6 × 10 ⁻¹			
		4.20††	π		9 × 10 ⁻⁸				
			π	0.39	6 × 10 ⁻⁶	1.3 × 10 ⁻¹			
KI	5.88	3.31§	π		4.4 × 10 ⁻⁶			0.132	9 × 10 ¹¹
		3.34††	π	0.34					
		3.31‡‡	π						
		3.31†††	π	0.34					
		3.04††	π						
RbI	5.73	3.02†††	π	0.34				0.018	
		4.13§	σ		2.3 × 10 ⁻⁹	3.8 × 10 ⁻²			
		4.15††	σ		~9 × 10 ⁻⁹				
		4.16‡‡	σ	0.34					
		3.09§	π	0.49	2.2 × 10 ⁻⁵	1.2 × 10 ⁻²			
CsI	5.30	2.30††	π		1.1 × 10 ⁻⁵				
		3.88§	σ	0.30	3.1 × 10 ⁻⁹	4.4 × 10 ⁻²			
		3.95††	σ		~9 × 10 ⁻⁹				
		3.09†††	π						
		2.31††	π						
		3.69	π	0.51	1.8 × 10 ⁻⁶	1.0			
		3.67§§§	π						
		4.28§§§	π						

† A single reference under a column heading indicates that all entries in that column are from that reference unless indicated otherwise. In general, values appearing to the right of a referenced luminescence peak energy come from the same reference unless indicated otherwise. All measurements except thermal quenching are at nominal liquid helium temperature.

‡ From Knox and Teegarden [59].

§ From Pooley and Runciman [54].

|| From Wasieła and Block [55].

||| From Blair *et al.* [60].

†† From Kabler and Patterson [50].

‡‡ From Ikezawa and Kojima [53].

§§ τ/τ_0 from [53] divided by τ from [50].

¶¶ From Williams and Kabler [61].

||| From Karasawa and Hirai [62].

††† From Nishimura [63].

‡‡‡ From Fishbach *et al.* [64].

§§§ From Falco *et al.* [65].

Table 2. Experimental parameters of self-trapped excitons in crystals other than alkali halides are listed, using a format similar to Table 1†

Crystal	STE emission peak (eV)	Width (eV)	Lifetime at LHeT τ_1 (s)	τ_2 (s)	τ_3 (s)	T_{quench} (10%) (K)
NH ₄ Cl	4.86‡	0.8	$5-10 \times 10^{-9}$			~90
NH ₄ Br	4.20‡	0.45	7.7×10^{-7}			~90
	4.2‡		$5-10 \times 10^{-9}$			~90
NH ₄ I	3.74‡	0.4				
MgF ₂	3.2	0.95	6.4×10^{-3}			70
CaF ₂	4.44§	1.1	8.3×10^{-3}	8.7×10^{-4}	8.8×10^{-3} ¶	80
SrF ₂	4.13§	1	5.9×10^{-3}	6.4×10^{-4}	7.7×10^{-3} ¶	70
BaF ₂	4.00§	0.9	5.3×10^{-3}	7.1×10^{-4}	5.4×10^{-3} ¶	
KMgF ₃	3.7††	1.4				
RbMgF ₃	2.9‡‡	0.8				≥100
AgCl	2.5§§	0.28	2×10^{-5}	1.5×10^{-4}	1.2×10^{-3}	60
SiO ₂	2.9¶¶-‡‡‡	0.8	10^{-3} †††, ‡‡‡			200†††, ‡‡‡

† In some of the crystals, the luminescence decay is represented as a sum of three lifetime components, listed here as

τ_1, τ_2, τ_3 .

‡ From Marrone and Kabler [58].

§ From Hayes and Stoneham [31].

¶ From Williams *et al.* [66].

|| From Williams *et al.* [56].

†† From Hayes *et al.* [57].

‡‡ From Koumvakalis and Sibley [17].

§§ From Marquardt *et al.* [21].

¶¶ From Hayes *et al.* [67].

||| From Itoh *et al.* [68].

††† From Sigel [69].

‡‡‡ From Tanimura *et al.* [70].

emission by unrelaxed excitons [33, 34, 88, 89]. Expressed as a fraction of the total self-trapped exciton luminescence (integrated over the band width), the efficiency of the free exciton luminescence

was found by Nishimura *et al.* to be 10^{-2} , 10^{-3} , 2×10^{-4} , and 6×10^{-4} in RbI, NaI, KI, and CsI, respectively [33]. These values were obtained for UV excitation just on the high energy side of the first exciton absorption band. Other workers using X-ray excitation obtained efficiencies one or two orders of magnitude smaller [34]. For X-ray excitation, free holes are produced, and they apparently self-trap quickly before many free excitons have a chance to form. Nishimura *et al.* also report that for UV excitation above the band gap, where free carriers would be produced, the free exciton luminescence is below their detection limit [33]. No alkali halides other than the iodides have shown detectable free exciton luminescence. However, in the condensed rare gases, coexistence of free excitons and self-trapped excitons is readily observed [26]. Refer to

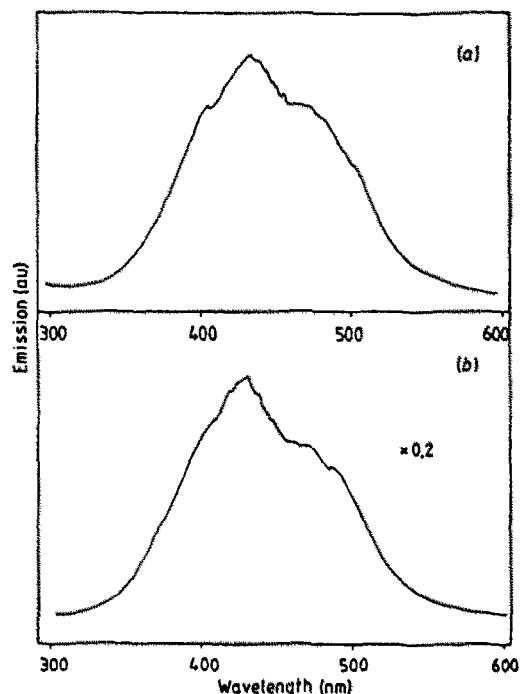


Fig. 8. X-Ray excited luminescence of quartz at (a) 77 K and (b) 1.7 K, attributed to self-trapped excitons (from Ref. 67).

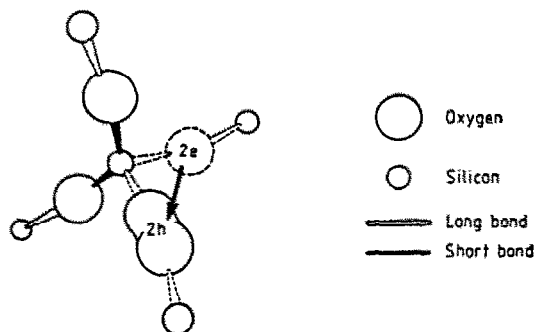


Fig. 9. Model proposed for the self-trapped exciton in SiO₂ (from Ref. 68).

Fig. 7(c) for luminescence of solid xenon, presented earlier.

The low fraction of free to self-trapped exciton luminescence found even for excitation in the first exciton peak of the most favorable case, RbI, suggests that the barrier-determined lifetime against self-trapping is around 100 ps, estimated as 10^{-2} of the allowed radiative lifetime of an unrelaxed exciton, assumed to be a few nanoseconds. For other alkali iodides, the lifetime against self-trapping is shorter, and for other alkali halides it is almost negligibly short, perhaps approaching 100 fs in absence of a barrier.

As the temperature is raised above 20 or 30 K, the free exciton luminescence in alkali iodides diminishes in favor of self-trapped excitons, consistent with thermal barriers against self-trapping which range from 11 to 33 meV [33, 34, 88, 89]. Several workers have also analyzed the dependence of self-trapped exciton yield on the concentration of defect quenching centers to obtain the temperature dependence of the range of free excitons [34, 88, 89]. The results are in good agreement with those based on free exciton luminescence.

AgCl and AgBr are an interesting pair of materials with regard to exciton self-trapping; the luminescence

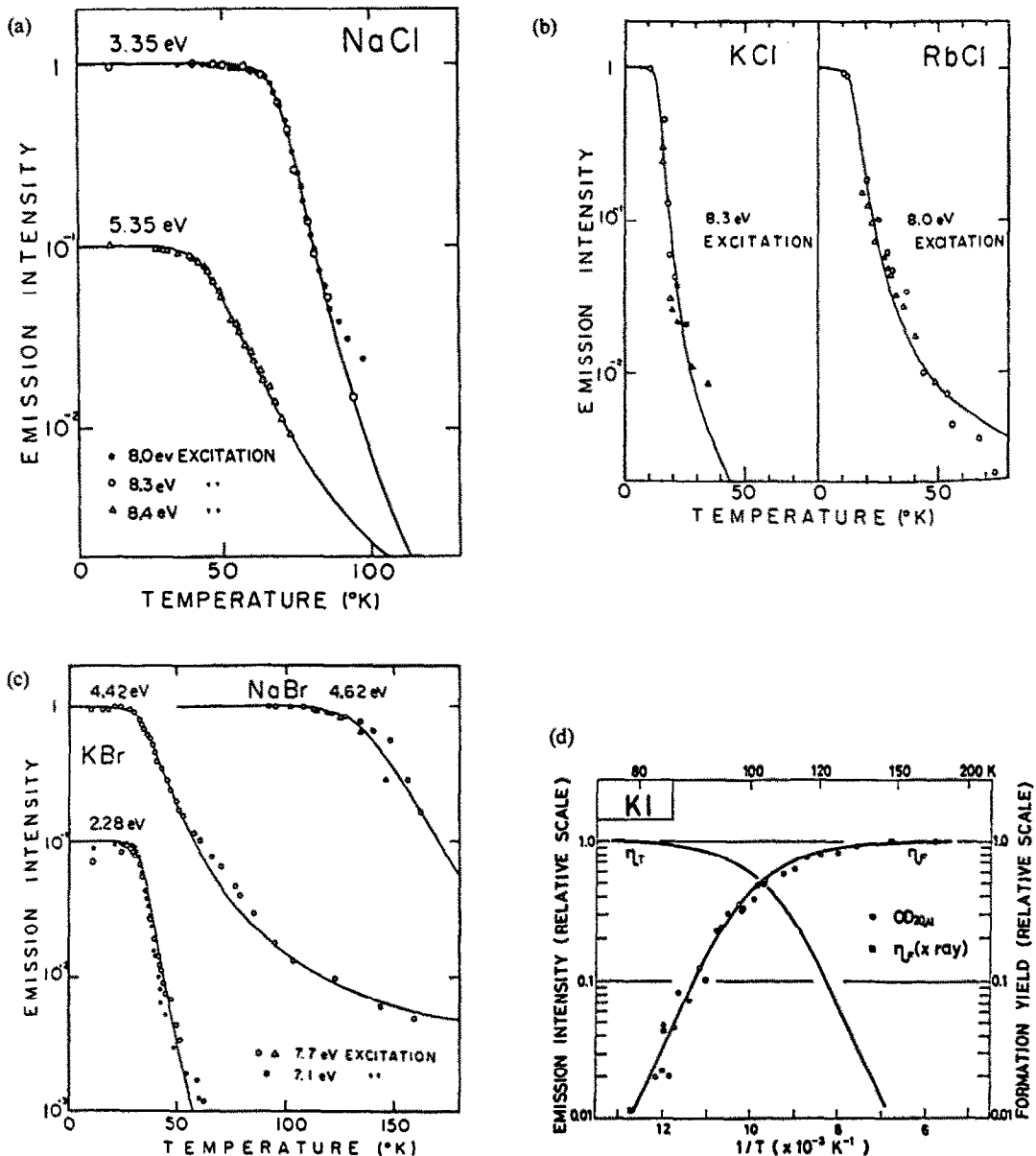


Fig. 10. Intensity of self-trapped exciton luminescence vs temperature. Parts (a)–(c) are from Ref. 53. Part (d) is from Ref. 79, and shows the rise of F-center absorption efficiency (η_F) correlated with the quenching of luminescence efficiency (η_T).

spectra indicate that excitons self-trap in AgCl but not in AgBr. As shown in Fig. 15, the luminescence spectrum of an $\text{AgCl}_x\text{Br}_{1-x}$ mixed crystal changes continuously from free-exciton-like with weak-coupling phonon side-bands to a broad self-trapped exciton emission band within the rather narrow composition range $0.4 < x < 0.5$. Kanzaki *et al.* showed that the exciton-phonon coupling strength in the mixed crystal could be tuned continuously and rather homogeneously by the composition in this system [90]. According to the theory of Toyozawa [91], self-trapping spontaneously occurs when the

exciton-phonon coupling strength exceeds a critical value, evidently corresponding to $x = 0.45$ in $\text{AgCl}_x\text{Br}_{1-x}$.

(F) Lifetime

Tables 1 and 2 summarize the luminescence decay times of self-trapped excitons at temperatures below the onset of thermal quenching. The π -polarized luminescence bands have lifetimes in the microsecond to millisecond range. The lifetimes become progressively shorter as the halide component of the alkali halides becomes heavier. Kabler and Patterson

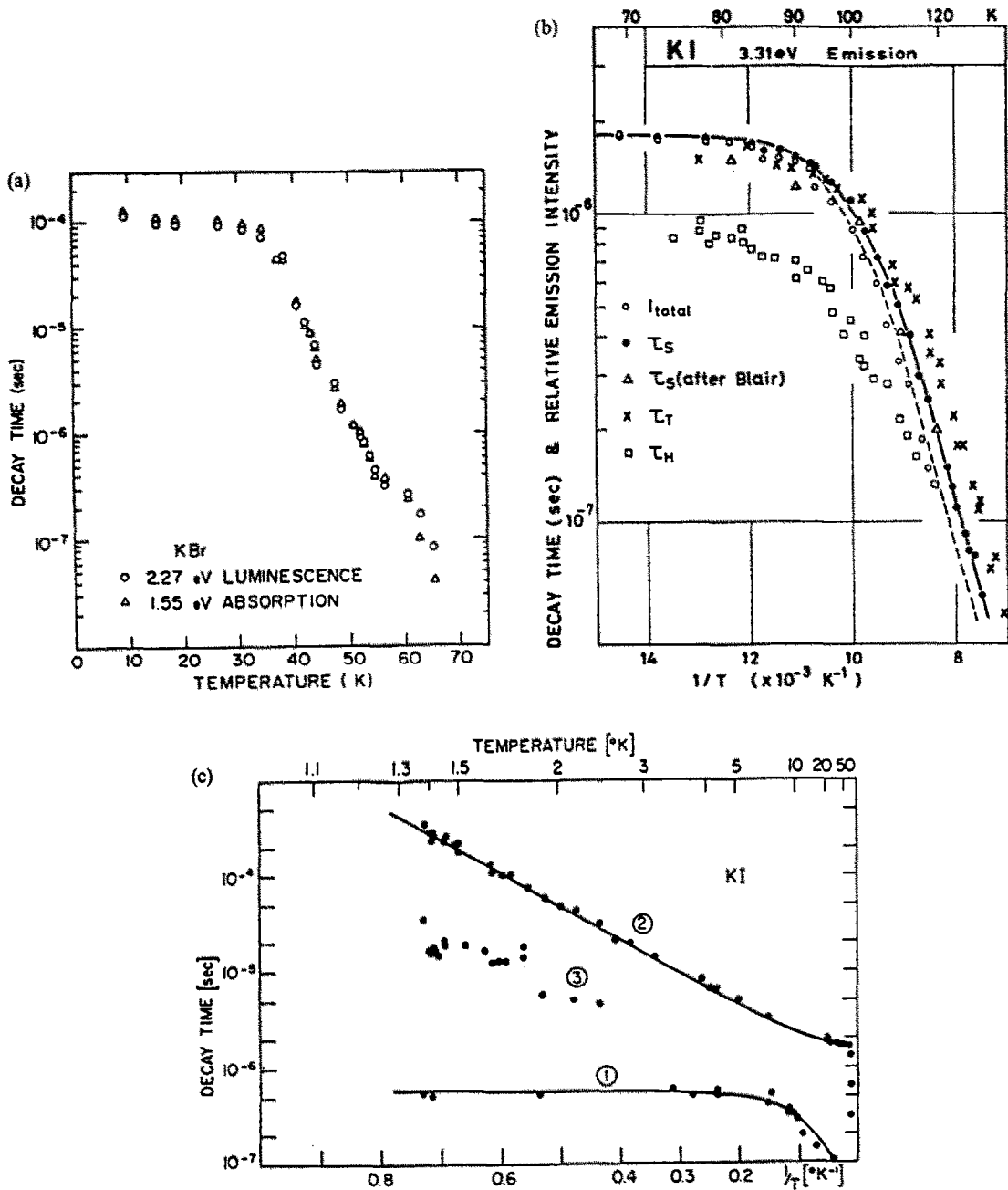
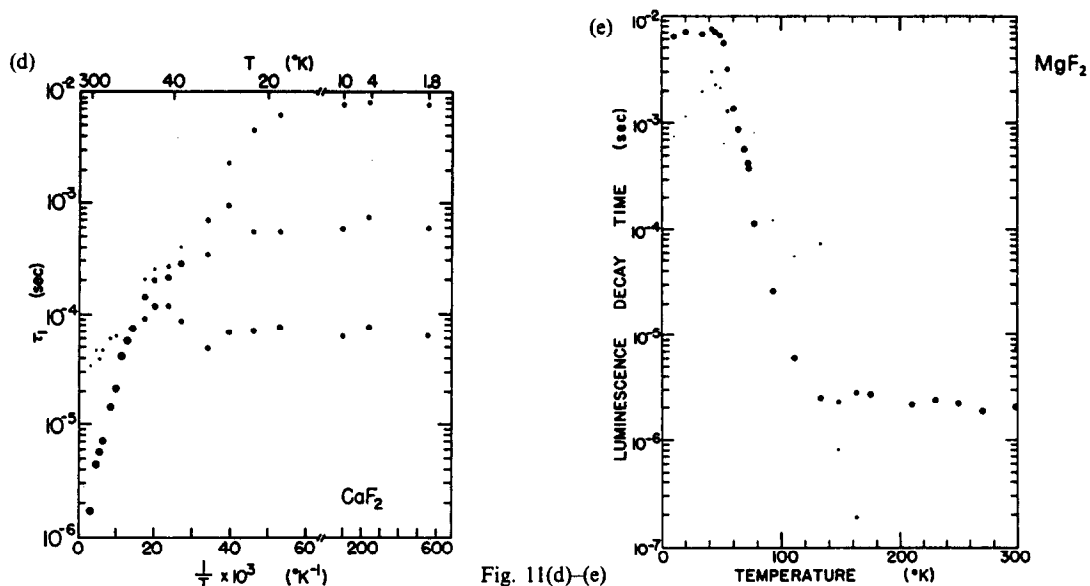


Fig. 11(a)-(c)



showed that the long-lived π luminescence originates from the lowest triplet state of the self-trapped exciton [50]. They showed that its π character and finite transition rate to the ground state comes from spin-orbit induced mixture of singlet Π character into the state denoted $^3\Sigma_u^+$.

As some of the alkali iodides are cooled below about 20 K, as many as three distinct lifetime compo-

nents can be observed in the triplet (π) luminescence bands. These are attributed to the zero-field-split triplet sublevels of the luminescent state, as discussed by Fishbach *et al.* [64]. Data for KI are shown in Fig. 11. In the alkaline earth fluorides, the luminescence from triplet self-trapped excitons also splits into three distinct lifetime components at low temperature [66], as shown in Fig. 11. The reader will appreciate

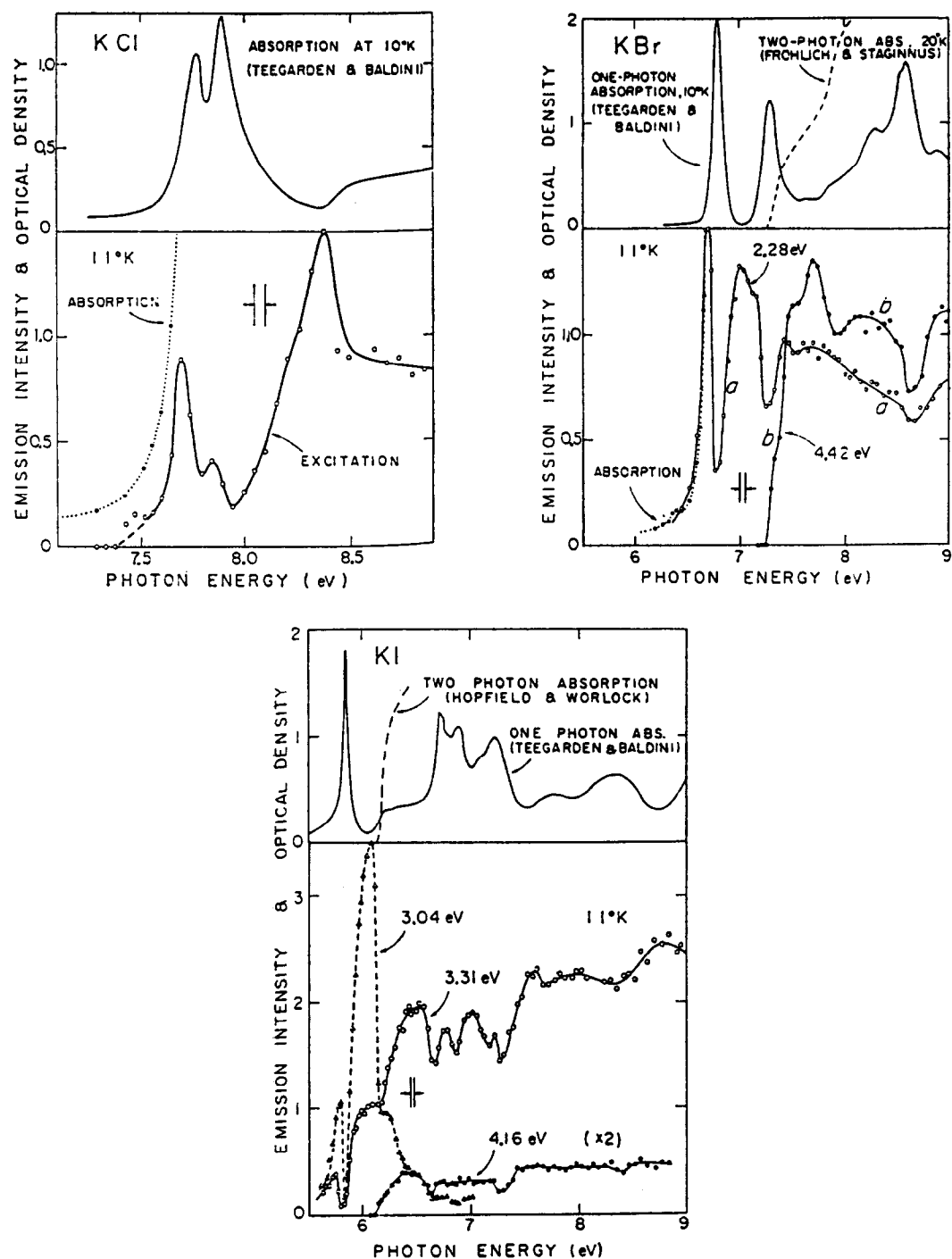


Fig. 13. Near-UV excitation spectra of self-trapped exciton luminescence in alkali halides at 11 K (from Ref. 53).

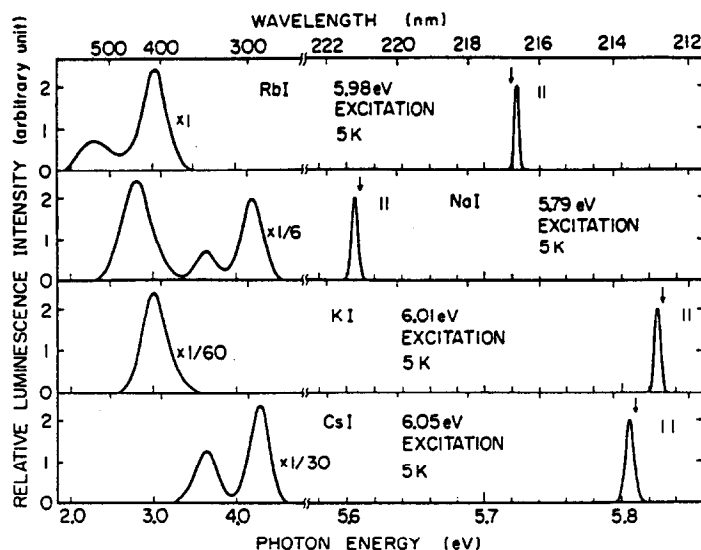


Fig. 14. Luminescence excited in the high-energy side of the first exciton absorption band at 5 K. The narrow peak on the right in each spectrum is due to free exciton radiative decay, while the broad bands are self-trapped exciton luminescence. The peak of the exciton absorption band is indicated by an arrow (from Ref. 33).

that such qualitative changes in the luminescence with temperature make it difficult to achieve generality in Table 2. The entries should be read with awareness that the temperature dependence may be rather complicated.

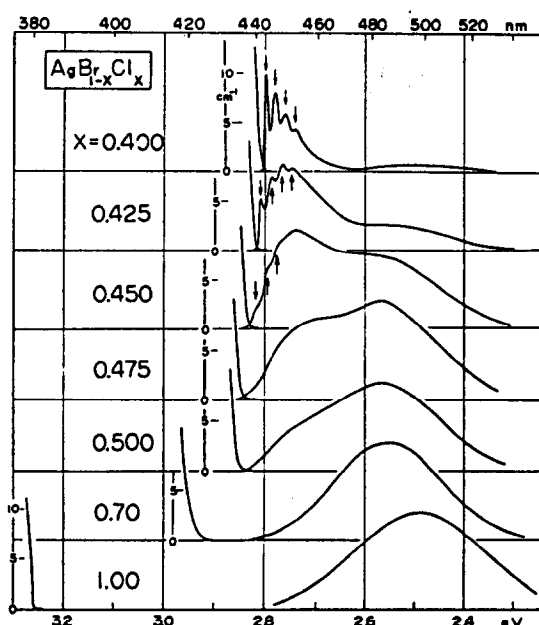
Lifetimes of the σ -polarized luminescence bands are generally between 2 and 10 ns, as listed in Table 1. This is clearly due to recombination in a singlet state. It is probably from a singlet associated with a

higher orbital level, but this has not been completely determined.

(G) Absorption in the excited state

While the self-trapped exciton exists, both the localized hole and excited electron can undergo optical excitation to still higher levels, yielding characteristic transient absorption spectra with decay times matching the STE luminescence decay [61, 92].

(a)



(b)

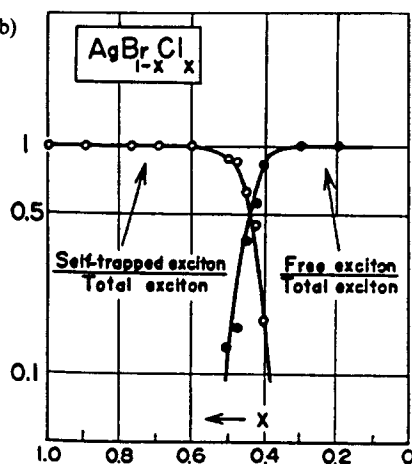


Fig. 15. Dependence of luminescence on composition, x , in $\text{AgBr}_{1-x}\text{Cl}_x$. Part (a) shows the changing spectrum, while (b) plots the fraction of luminescence which is broad-band (self-trapped exciton) and that which exhibits sharp-line vibronic structure (free exciton) (from Ref. 90).

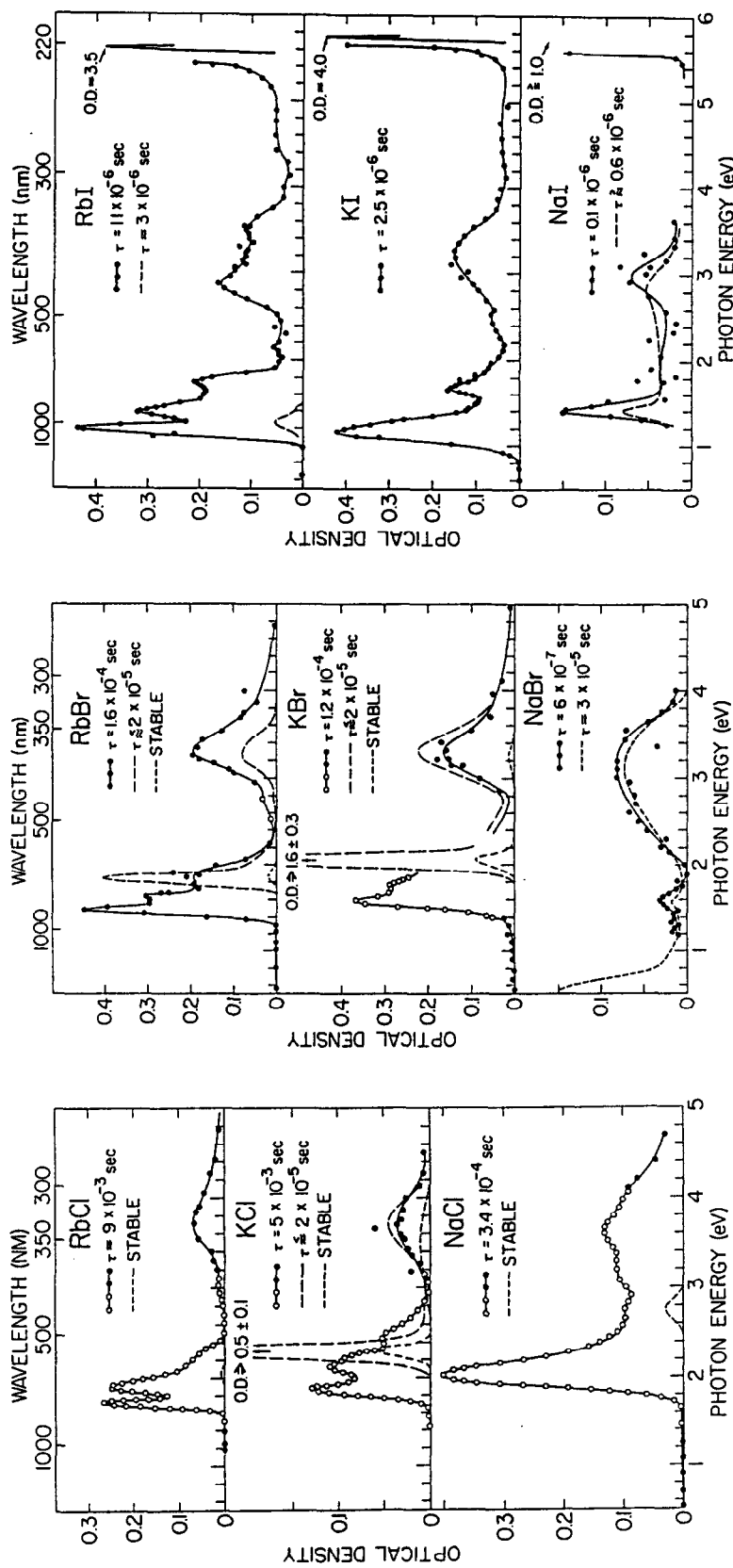


Fig. 16. Transient absorption spectra of self-trapped excitons in alkali halides following electron pulse excitation at 9 K. Components marked "stable", and their associated transient replicas (dashed curves), are due to F centers and H centers (from Ref. 61).

Spectra of self-trapped excitons in alkali halides, alkaline earth halides, silver chloride, silicon dioxide, and condensed neon are shown in Figs 16–18.

The absorption bands in alkali and alkaline earth halides divide naturally into two groups. A broad ultraviolet band resembling both the self-trapped hole (V_k) spectrum and the interstitial halogen (H center) spectrum was shown to be the $\sigma_u \rightarrow \sigma_g$ hole transition in the halogen molecular ion model of the STE, as illustrated in Figs 2 and 4. That is, the hole is excited from the top-most σ_u orbital to the deeper σ_g orbital, while the excited electron remains relatively unaffected.

At longer wavelength, usually bracketing the F-band in alkali and alkaline earth halides, is a sequence of transitions of the electron component of the STE to successively higher orbitals around the self-trapped hole, finally terminating in an ionization limit where the electron escapes to leave behind a V_k center. The optical binding energy of the electron in the STE is in the range 1.1–2.8 eV in different alkali halides, from 2 to 3 times the free exciton binding energy, and similar to the F-center binding energy in the same crystal. In the alkaline earth fluorides [66], the optical binding energy of the STE electron is 2–3.5 eV, again

comparable to that of the F center. The self-trapped exciton spectrum in AgCl resembles that of the self-trapped hole, comparing Figs 18(a) and 3(d). Transient absorption spectra of self-trapped excitons involving halogen impurities in mixed halide crystals have been reported by Arimoto *et al.* [93]. In Fig. 18(c) two transient absorption bands are seen in condensed neon after electron pulse excitation. The peak labeled “a” near 1.92 eV is due to the “atomic” or one-center self-trapped exciton, while the broader band labeled “m” is due to the “molecular” STE existing as a dimer Ne_2^+ . Both peaks shift progressively in time as the bubble size increases by vacancy capture. The “a” and “m” STE configurations are illustrated in Fig. 18(d).

(H) Double excitation spectroscopy

It is important to be able to identify the absorption bands in the excited electron series of Figs 16–18 with specific transitions of the STE. The absorption spectra alone do not provide sufficient information for identification. Based on an analogy to the M center in alkali halides, it might seem reasonable to assign the lowest energy electron transition to $1s\sigma_g \rightarrow 2p\sigma_u$, as shown on the left side of Fig. 4(a).

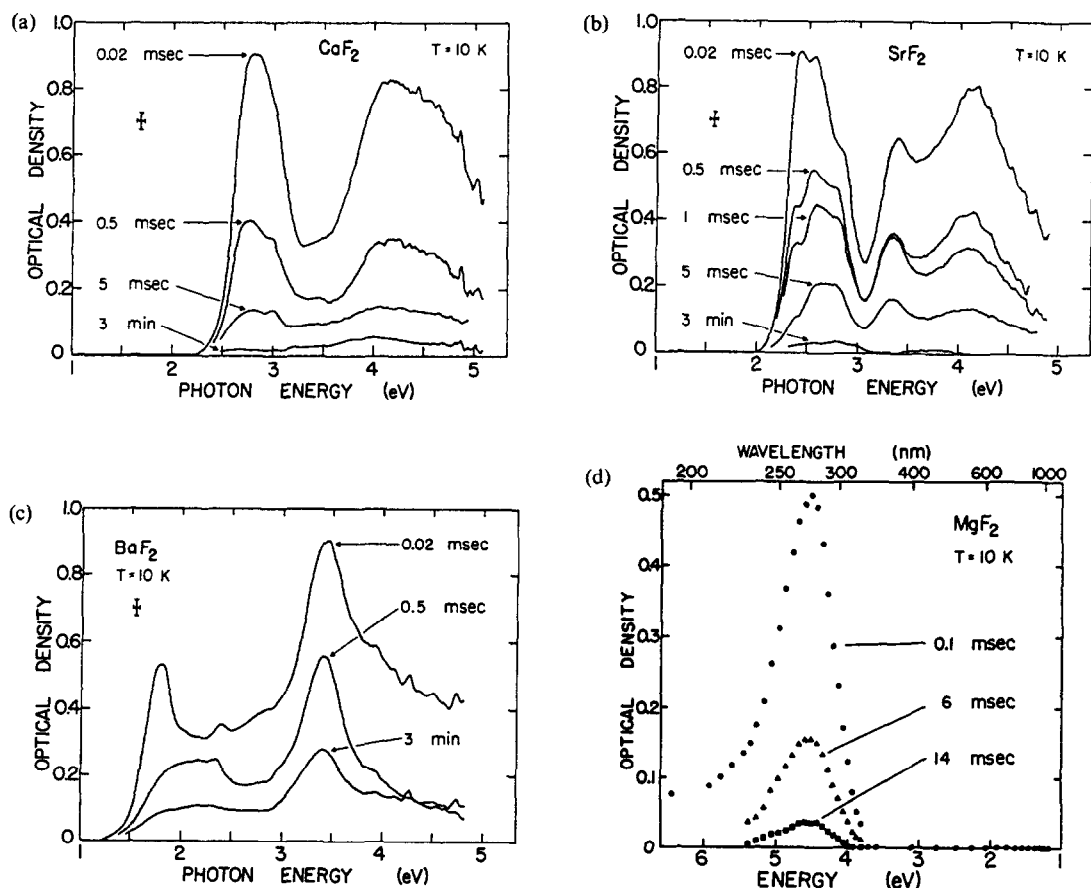


Fig. 17. Transient absorption spectra of self-trapped excitons in alkaline earth fluorides following electron pulse excitation at 10 K. Parts (a)–(c) are from Ref. 66 and (d) from Ref. 56.

Calculations based on the $(V_k + e)$ model of the self-trapped exciton indicated that the $1s\sigma_g \rightarrow 2p\sigma_u$ electron transition ($a_{1g} \rightarrow b_{3u}$) should lie substantially lower than the observed lowest-energy absorption band [94, 95]. The first indication that a different assignment was in order came from double-excitation experiments in 1975 [96]. An electron pulse created a STE population, which after a few microseconds settled into a population occupying the lowest triplet state. Then a ruby laser pulse promoted essentially

all of the self-trapped exciton population to higher states. This caused crossing from triplet to singlet manifolds in the excited state, as well as enhanced F center production in some crystals. Using polarized exciting light and detecting polarized luminescence, it was found that the lowest STE absorption band in NaCl is π polarized with respect to the symmetry axis of the self-trapped hole, inconsistent with the $1s\sigma_g \rightarrow 2p\sigma_u$ assignment for the electron transition assumed earlier. Tanimura and co-workers used an

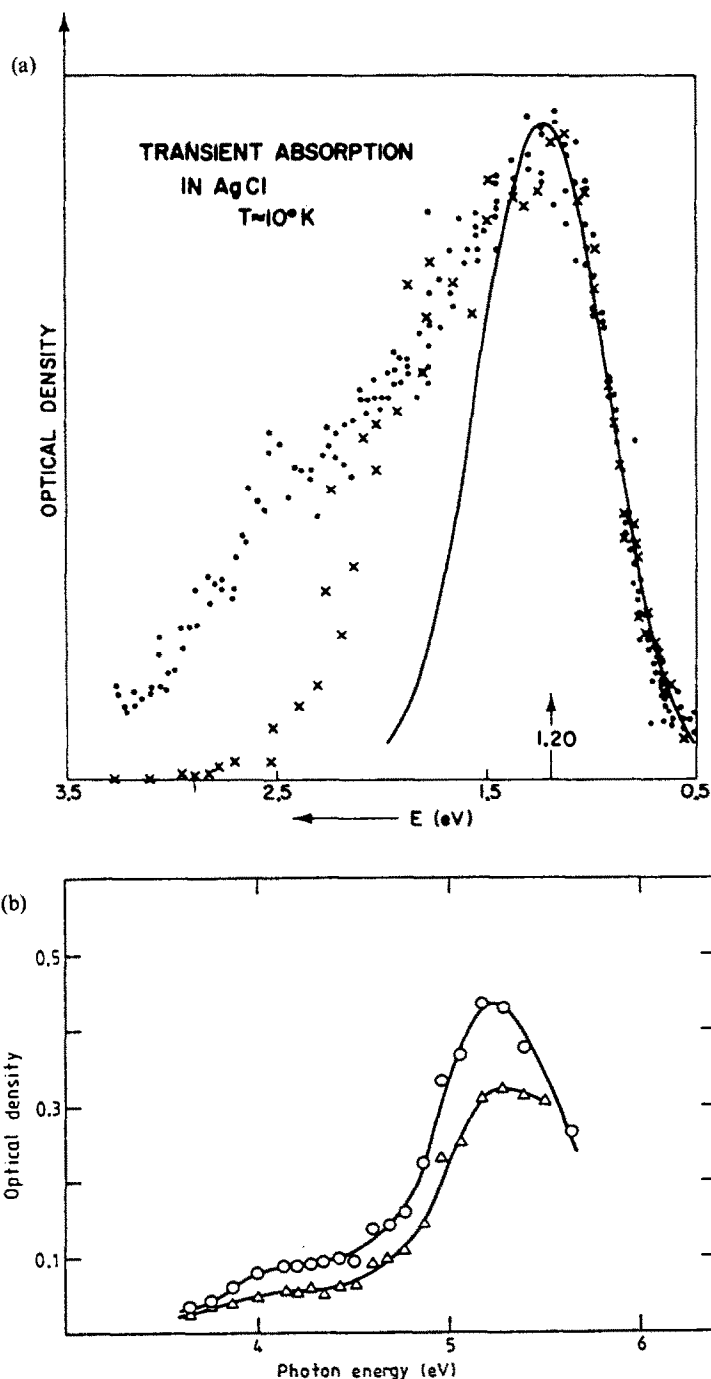


Fig. 18(a)-(b)

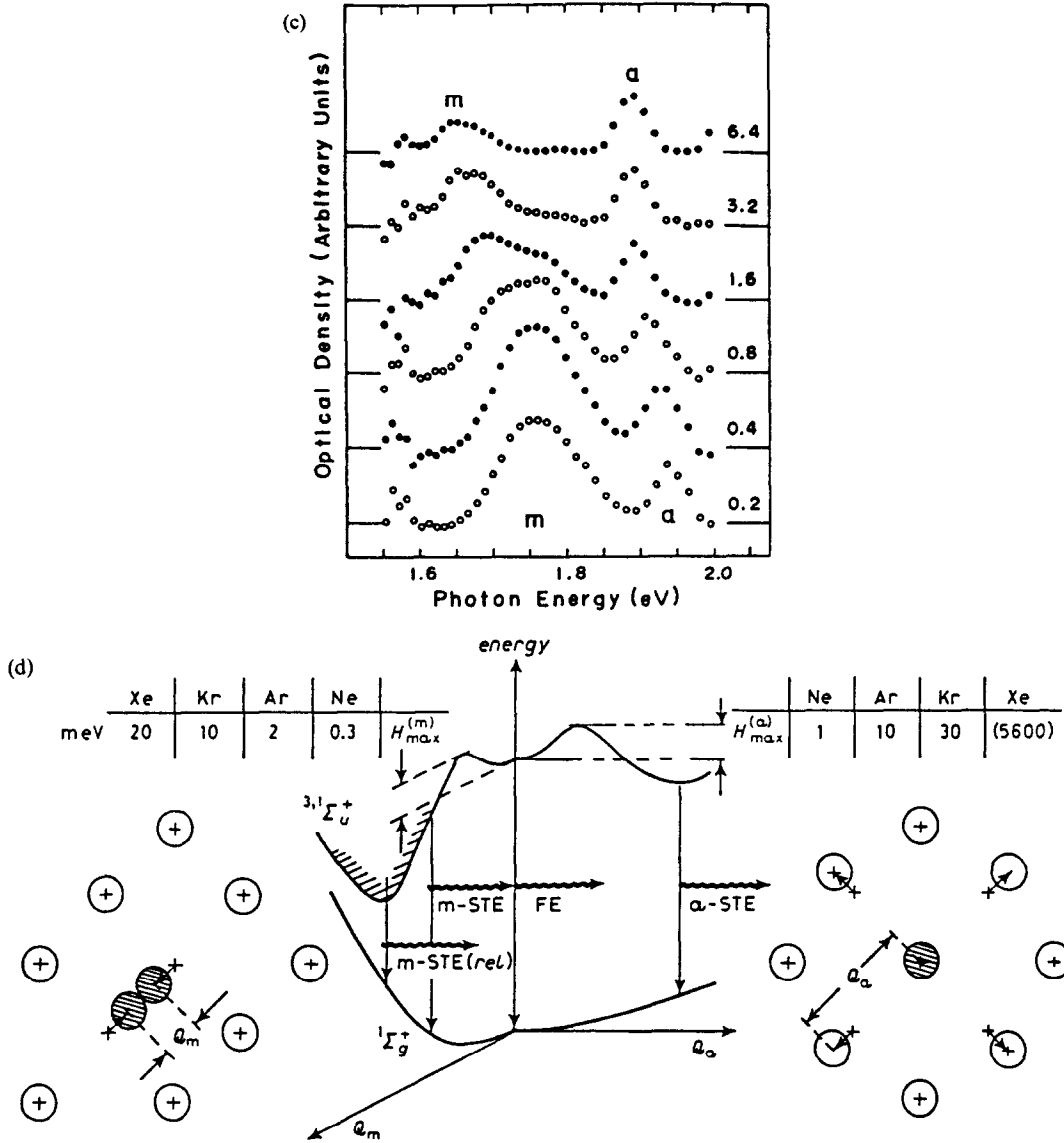


Fig. 18(c)-(d)

Fig. 18. Transient absorption spectra in (a) AgCl, (b) SiO₂, and (c) condensed neon following electron pulse irradiation. The 1.2-eV band and poorly-resolved absorption near 1.8 eV in AgCl [21] are similar to self-trapped hole absorption [Fig. 3(d)], but are transient and thus are attributed to self-trapped excitons. The transient absorption spectrum in crystalline (Δ) and amorphous ($\circ$) SiO₂ (from Ref. 68) has been tentatively attributed to self-trapped excitons of a structure as suggested in Fig. 9. Absorption spectra of atomic (*a*) and molecular (*m*) self-trapped excitons in condensed neon are shown at 0.2, 0.4, 0.8, 1.6, 3.2, and 6.4 μ s after electron-pulse excitation (Ref. 93). Schematic structures and potential curves of the (*a*) and (*m*) self-trapped excitons are shown in (d) (from Zimmerer, Ref. 26).

electron pulse and tunable dye laser combination to make thorough studies by this technique [97, 98]. They showed that the lowest energy STE absorption band is often π polarized, and furthermore they deconvoluted the first three STE excited states from the overlapping spectral components. The peaks assigned by Tanimura *et al.* in the STE absorption spectra for KCl and KBr are shown in Fig. 19.

(I) Ultrafast spectroscopy

Two-photon absorption of near-ultraviolet laser pulses can provide the necessary 6–12 eV band-gap

excitation needed to create electron–hole pairs in halide crystals. Thus the capabilities of ultrafast laser spectroscopy are available for studies of self-trapped excitons and their subsequent transformations, e.g. lattice defect formation. The first observation of this kind used a 20-ps pulse of 266 nm light to create electron–hole pairs in KCl by two-photon absorption. Time-dependent absorption measured near the peak of the F center absorption band by 30-ps, 532 nm pulses showed that F centers were formed in their ground state within 10 ps of e – h pair generation [99]. As is discussed by Itoh and Tanimura in this

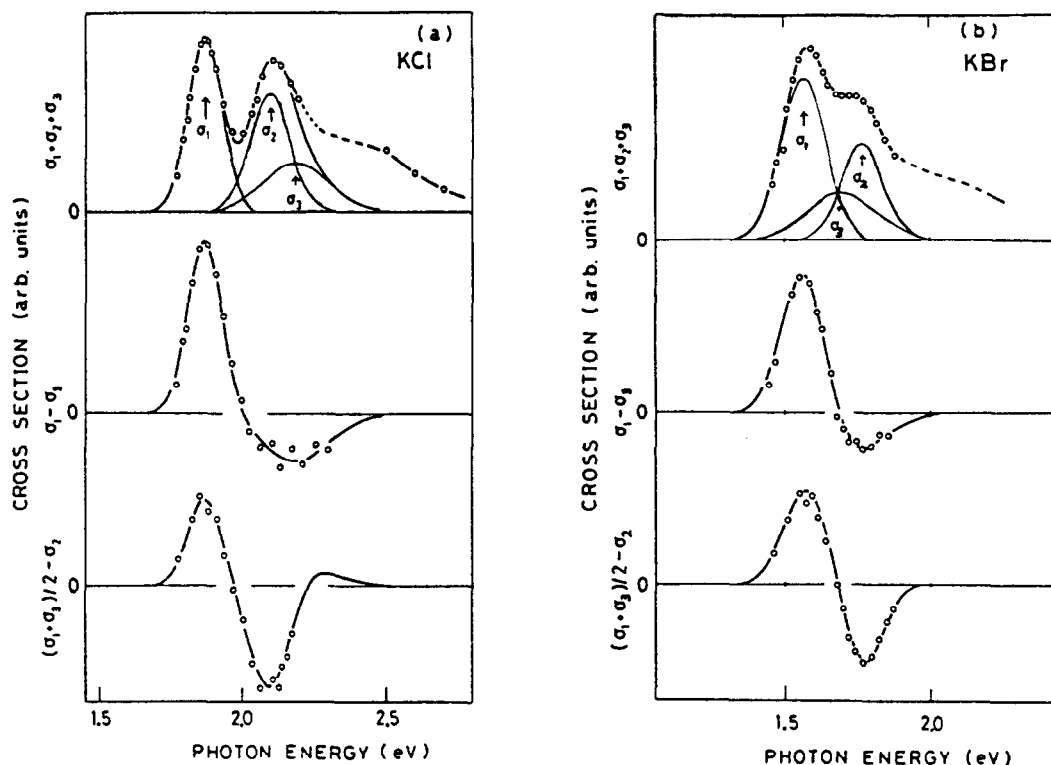


Fig. 19. Decomposition of the self-trapped exciton absorption spectra due to electron transitions in (a) KCl and (b) KBr (see Fig. 16) into three specific transitions labeled by cross-sections σ_1 , σ_2 , and σ_3 . In terms of the electron orbitals of Fig. 4(a) for D_{2h} symmetry, σ_1 is $a_{1g} \rightarrow b_{2u}$, σ_2 is $a_{1g} \rightarrow b_{1u}$, and σ_3 is $a_{1g} \rightarrow b_{3u}$. The decomposition is based on polarization measurements as shown in the lower parts of the figures. The analysis is discussed in Ref. 98, from which these figures are taken.

issue, photochemical F center formation is known to proceed through states of the self-trapped exciton in alkali halides. Thus the F center formation time sets an upper limit on the time for exciton self-trapping in alkali halides, though not necessarily on the time for STEs to reach their lowest excited state. Subsequent spectroscopy with 40-ps white light pulses as the probe were able to monitor the self-trapped exciton absorption bands in KI directly, as well as the self-trapped hole (V_k) band [40]. It was concluded that hole self-trapping proceeds as fast as can be determined by deconvolution of the 40-ps measurement pulse. However, a highly excited electron at least above the $2p\sigma_u$ orbital of the STE in KI requires 100 ps to reach the lowest excited electron orbital, $1s\sigma_g$ [100]. Similarly slow STE relaxation is found for RbI, but the relaxation seems very fast for lighter halides.

Recently, Williams *et al.* have used 2-ps pulses for spectroscopy of self-trapping and F center formation [42, 43]. These measurements are being extended to the subpicosecond regime. Results obtained with 2-ps pulses in KBr at room temperature are shown in Figs 20 and 21. The spectra may be compared with Fig. 16. The F center peak is readily identifiable at 620 nm. We believe the absorption around 700 nm is part of the STE electron absorption bands. The measurements in Fig. 21 show that self-trapped

excitons and about half of the F centers are both formed within 2 ps after $e-h$ pair generation.

(J) Electron paramagnetic resonance

Electron paramagnetic resonance has been the most powerful tool for detailed study of the V_k center, i.e. the self-trapped hole rendered stable against recombination by trapping the electron elsewhere. Although the STE is a transient species, EPR spectroscopy can be performed by the method of optical detection. A crystal under continual X-ray excitation to produce self-trapped excitons is placed in a magnetic field and microwave cavity. The STE luminescence and its linear or circular polarization is detected at sufficiently low temperature that the equilibrium population distribution among the magnetic field-split sublevels in the crystal field distinctly favors the lower levels. When the microwave frequency matches a magnetically-tuned resonance of the STE, the polarization and total intensity of the luminescence are altered by resonance transitions out of the equilibrium population distribution. Schematic illustrations of the sublevel splittings in applied magnetic fields are shown as inserts in Fig. 22. Optically-detected EPR spectra of self-trapped excitons in KBr [51], CsBr [51], and KCl [101] are also shown in Fig. 22. The seven-line hyperfine patterns for KBr

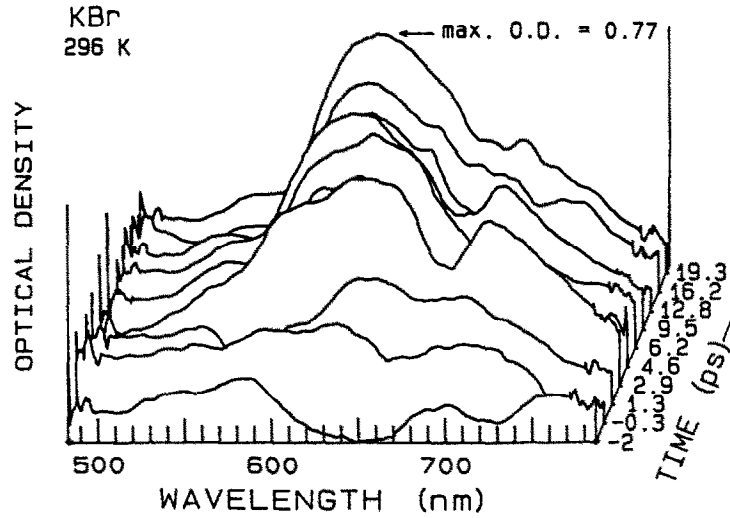


Fig. 20. Evolution of F-band and self-trapped exciton absorption measured with 2-ps pulses in KBr at 296 K following electron-hole pair generation by two-photon absorption (from Ref. 43).

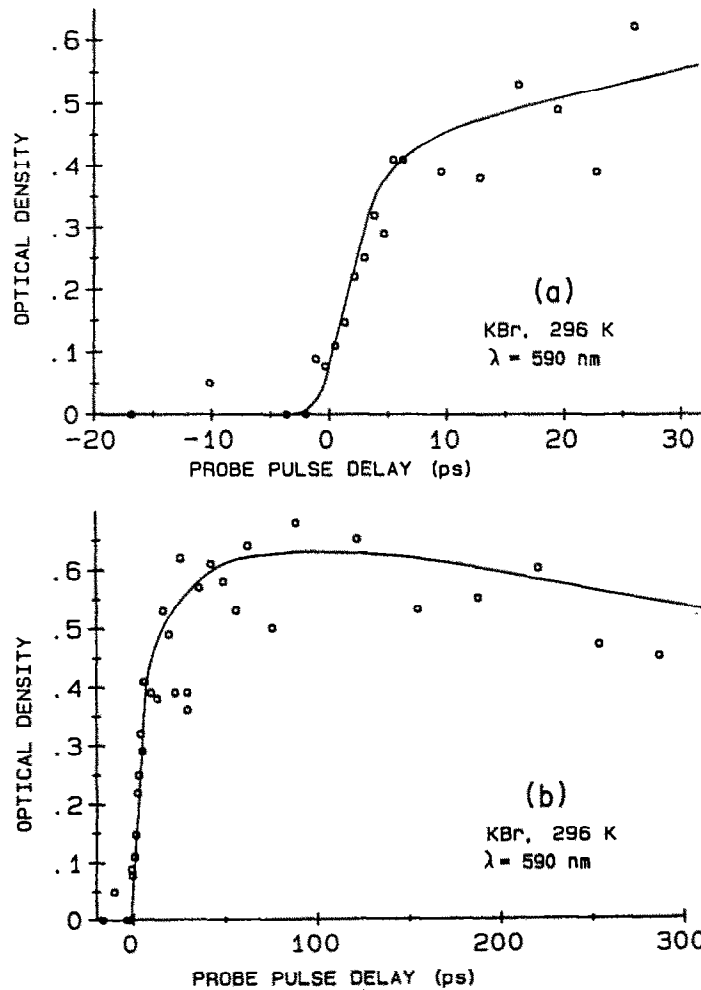


Fig. 21. Optical density at 590 nm in KBr at 296 K as a function of delay after two-photon band-gap excitation by a 1.5-ps, 297-nm pulse. The solid curves are fits to the convolution function described in Ref. 41, where the generation function is

$$K(t-t') = \{C_1[1 - \exp(-(t-t')/\tau_1)] + C_2[1 - \exp(-(t-t')/\tau_2)]\} \exp(-(t-t')/\tau_3)$$

with $\tau_1 = 2$ ps, $\tau_2 = 50$ ps, $\tau_3 = 800$ ps, and $C_1 = C_2 = 0.37$ (from Ref. 43). The data are shown for two time ranges in (a) and (b).

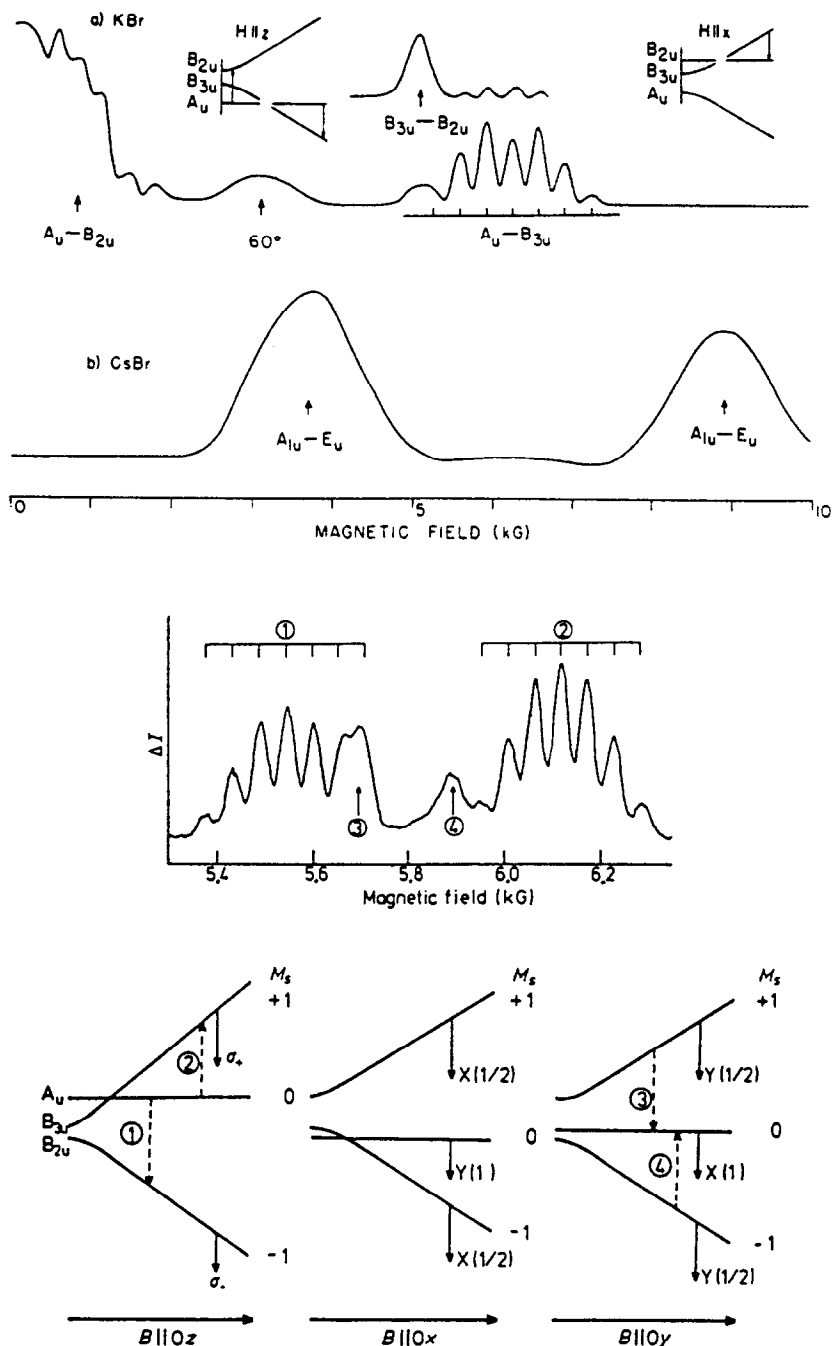


Fig. 22. Optically-detected EPR spectra of self-trapped excitons in KBr and CsBr (from Ref. 51) and in KCl (from Ref. 101). Also shown are the triplet energy levels of the self-trapped exciton in a magnetic field, indicating the luminescence and microwave transitions that can occur.

and KCl are similar to the well-known EPR spectra of stable self-trapped holes (V_k centers) in these materials. The V_k hyperfine pattern arises from interaction of the electron (hole) spin with two equivalent Br or Cl nuclei, each having nuclear spin $I = 3/2$. The optically-detected hyperfine peaks for the self-trapped exciton are substantially broader than are the hyperfine peaks of the V_k center [9, 10]. They are in

fact comparable with F-center EPR linewidths, suggesting similar extents of the electron wavefunctions of the self-trapped exciton and the F center. In KCl, the width of each STE hyperfine component is 38 G, compared with the F-center linewidth of 55 G and the V_k hyperfine linewidth of 1.4 G [10]. EPR parameters for self-trapped excitons in alkali halides have been summarized by Itoh [52].

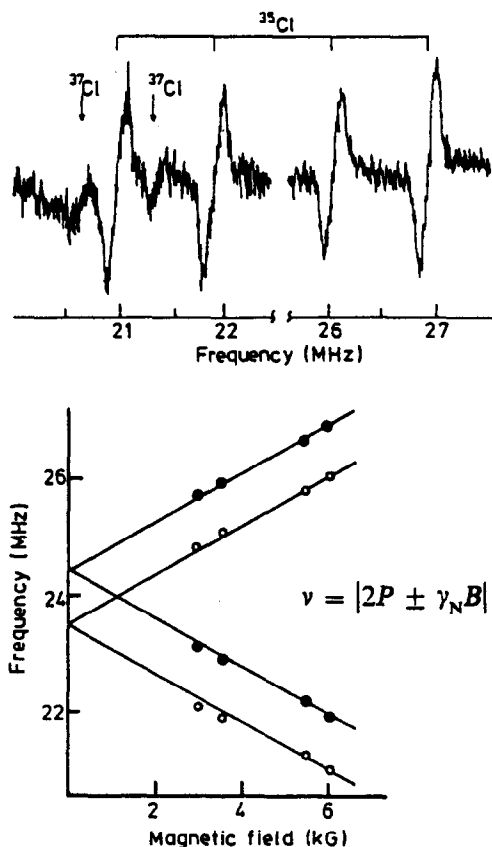


Fig. 23. Nuclear quadrupole resonance lines in the ENDOR spectrum of self-trapped excitons in KCl are shown, along with the magnetic field dependence of the lines for $B \parallel (110)$ (from Ref. 101).

(K) ENDOR and nuclear quadrupole resonance

Block *et al.* measured optically-detected ENDOR of the STE in KCl [101]. Comparison of the contact interactions on nuclei surrounding the STE and the F center indicated that the radial extent of the wavefunction of the bound electron is similar to that of the F center. Also of interest was a set of four lines due to quadrupole resonance of the halogen nuclei. The spectrum and its magnetic field dependence are shown in Fig. 23. The frequencies of the lines depend on the magnetic field as

$$\nu = |2P \pm \gamma_N B|, \quad (5)$$

where γ_N is the gyromagnetic ratio of the ^{35}Cl nucleus and $2P$ is the quadrupole splitting between the $\pm 3/2$ and $\pm 1/2$ nuclear spin sublevels. Figure 23 shows the existence of two sets of two lines, occurring with equal probability and characterized by two different values, aP and bP , of the quadrupole interaction. The existence of two values of the quadrupole interaction shows that the central nuclei of the STE are not equivalent. The authors concluded that the only reasonable explanation is that the self-trapped exciton relaxes off-center along a (110) direction, lowering the symmetry from D_{2h} to C_{2v} [101].

(L) Self-trapped exciton formation by F–H recombination

It has been shown by Purdy and Murray for KCl [81] and by Tanimura and Okada for KBr [102] that recombination of H centers and F centers during low temperature annealing produces π luminescence of self-trapped excitons. The sigma luminescence of STEs is not produced directly by annealing of F–H pairs, as shown in the 1980 paper of Tanimura and Okada [102].

The STE π luminescent state and the F–H defect pair lie on the same adiabatic potential surface [103]. That is, one is transformed into the other by H center motion relative to the F center. The demonstration of STE π thermoluminescence from annealing of F–H pairs is an important demonstration that in at least KCl [81] and KBr [102], the F–H pair energy is no more than about 70 meV lower than the triplet STE luminescent state. Coupling this fact with the observation of thermally-activated conversion from the STE π luminescent state to F–H pairs [54, 74, 78], we are led to conclude that in general, the F–H pair and the STE π luminescent state lie very close to the same energy on the same adiabatic potential surface, plotted as a function of F–H pair separation [75–78]. Furthermore, the activation barriers are modest, corresponding approximately to H center migration energies for both the forward and backward reaction.

(M) Hopping transport of self-trapped excitons

Coherent transport of free excitons in alkali iodides at low temperature has been discussed earlier. Hopping transport of V_k centers was discussed in our review of properties of the self-trapped holes. The measurements were straightforward because the V_k centers are essentially stable defects until the crystal is warmed to activate their motion. Measuring the hopping transport of self-trapped excitons is more challenging, and only a few such measurements have been reported. Vasil'chenko *et al.* measured the temperature dependence of Ag^+ luminescence excited in the exciton absorption bands of the host NaCl crystal [104]. From it they deduced an activation energy for the motion of the self-trapped exciton of 0.21 eV, to be compared with 0.44 eV for the motion of self-trapped holes in NaCl:Ag. Tanimura and Itoh conducted similar experiments on NaCl:Li excited by X-rays and electron irradiation [105]. They deduced an activation energy of 0.15 eV for self-trapped exciton hopping transport. NaCl appears to be the only crystal for which the activation energy has been measured. It was proposed by both groups that the presence of the excited electron may moderate polar relaxation around the V_k center and thus assist in diffusion. Earlier, Song had applied small polaron theory to the hopping diffusion of V_k centers and of self-trapped excitons ($V_k + e$ model) in alkali halides, finding that the diffusion energy of the $V_k + e$ should be smaller than that of the V_k center by a factor of 2–3 [106]. The present authors have

speculated on how off-center self-trapped excitons may diffuse [107].

If the STE in alkali halides is effectively a nearest-neighbor F-H pair, the mechanism of the hopping motion should be different from that of the V_k center. Recent work by Song and Chen [108] suggests that the nearest-neighbor F-H pair, or the off-center STE, moves around the lattice through a combination of exchange of sites between the "F" and "H" centers and rotation of the "H" center. There are two groups of alkali halides in this context. The first group is characterized by small activation energy for the F-H exchange motion (about 0.2 eV) and smaller off-center shift of the V_k core, therefore smaller F-H separation. This group includes NaCl, NaBr, and presumably the iodides. In the other group, characterized by large F-H separation in the STE state, the energy required to accomplish F-H exchange is much larger, about 1 eV. KCl and KBr belong to the second group. This division seems to be in agreement with experimental data available so far; for example, impurity excitation by STE migration was not found in KCl:Ti and RbCl:Ti [104].

5. THEORY OF SELF-TRAPPED EXCITONS

There have been two major avenues of approach to the theory of self-trapped excitons. On the one hand, the transition from free exciton to localized state has been treated by Toyozawa [5]. This approach draws in part from the general problem of small polaron formation [1-3]. The results compare most readily with experiments on transport.

Higashimura *et al.* have studied the problem of the V_k center as well as that of the STE in alkali halides using a continuum approach [109]. The electron is treated in the effective mass approximation and the electron-lattice interaction by the Fröhlich Hamiltonian. Within the $V_k + e$ model, they report reasonable agreement with absorption and emission energies.

In the second major approach, the structure of the STE has been addressed using theoretical methods developed for defects in solids, i.e. treating the STE as a transient lattice defect, and calculating the electronic structure and ion positions. This approach gives the most detailed parameters for comparison to spectroscopy, and is the subject of the following discussion.

In 1974, Stoneham presented the first such calculation of the STE in NaCl [110]. It was based on a computer code (ATMOL) which performed a Hartree-Fock calculation on a small cluster of ions embedded in an ionic lattice of point charges. The cluster comprised a pair of Na^+ and a pair of Cl^- ions, the rest of the lattice being represented by point charges. The lattice was not allowed to relax during the calculation, but effects of fixed amounts of ionic displacement were studied. This gave fairly good

agreement with observations regarding the hole excited states. The agreement was not as satisfactory when the electron excited states were involved. This was attributed to the inadequacy of the excited state basis and the small size of the cluster.

Next, sustained efforts were made to determine the electron excited states using the ion-size approximation [94, 95]. This method was developed earlier by Bartram *et al.* [111] as a simplified and efficient version of the extended-ion method, and gave good results in the study of F centers. Its basic assumption is that the excited electron pseudo-wavefunction varies slowly over the space occupied by the core and valence electrons.

Before summarizing the results obtained by this method, it is worthwhile to describe some aspects of this work. In the studies of F centers, the ion-size terms contributed relatively small corrections to the energies of the electron obtained with the point-ion approximation. With and without the ion-size terms, the F center electron was found to be deeply bound. The STE was studied with the V_k core at the known V_k position in the lattice, while the other ions were at their perfect lattice sites, or were given an arbitrary displacement. In particular, displacements of the closest pair of cations which appear to inhibit translation of the V_k center were explored. When the ion-size terms were absent, Song *et al.* always found binding states for the s -like electron orbitals (a_{1g} symmetry in the D_{2h} group of the V_k center) using a single optimized basis [95]. Adding extra basis functions did not noticeably change the energy. When the ion-size terms were added, the situation was very different. The single optimized basis often turned out to be very diffuse and its energy very shallow [112]. It was often necessary to use a large basis (about a dozen functions) to obtain the a_{1g} energy of about -3 eV, for the state with the excited electron in the lowest orbital of a_{1g} symmetry, corresponding to the π -luminescent state. This is an interesting point to come back to when we discuss the overall situation of the STE later.

In the ion-size (or extended-ion) method, we do not have access to the hole states. The electron excited states b_{1u} , b_{2u} , b_{3u} , and a_{1g}^* (next higher electron orbital of a_{1g} symmetry) were obtained for the fluorides and chlorides [95] and also for potassium halides [94]. They consistently show fairly good agreement for two of the s -like to p -like transitions, $a_{1g} \rightarrow b_{1u}$, b_{2u} , with transition energies around 2 eV. The ion-size method also gave the a_{1g}^* excited state at about the right place to be the initial state for the σ emission band.

There were two points which remained unsatisfactory in those treatments. For all alkali halides, Song *et al.* found the b_{3u} (p -like) excited electron orbital too low (giving only about 1 eV for the $a_{1g} \rightarrow b_{3u}$ transition energy), and in fact the lowest of the b_u states [94, 95]. Later, measurements of the polarization of the corresponding absorption bands showed

that $a_{1g} \rightarrow b_{3u}$ is often the highest energy transition [98]. This was obviously a weak point in the theory. The second unsatisfactory aspect was in the spin density of the excited electron of the STE. Ion-size calculations gave a spin density which had a very strongly anisotropic distribution, while a later ENDOR study [101] gave an almost isotropic distribution with the absolute value quite well correlated to that of the F center.

In 1977, Itoh *et al.* presented new theoretical studies on the structure of the STE in NaCl [113]. Their method consisted of combining the semi-empirical CNDO method with the HADES code which was developed earlier to study defects in ionic crystals. In the CNDO method, a small number of empirically fitted parameters are used. Under a set of simplifying approximations, the method simulates a many-electron Hartree-Fock calculation. A possible shortcoming is that the basis may be too restricted. Nevertheless, the method is frequently used with varying success in quantum chemistry and solid state physics. The HADES code was used to determine the lattice relaxation in the presence of an impurity atom or molecule using the shell model. It has been used with considerable success to study such quantities as the formation energy or migration energy of defects in ionic crystals [114].

Itoh *et al.* first used the CNDO method alone to determine the spectrum of excited states of the STE in NaCl [113]. They obtained agreement with data for the hole excited states and very analogous results to the ion-size calculation [94, 95] regarding the electron excited states. Notably, they found the $a_{1g} \rightarrow b_{3u}$ transition at much lower energy than experiment showed. All these treatments were done on the configuration of the STE in D_{2h} symmetry, i.e. a V_k center with a bound electron. It is therefore noteworthy that three works [94, 95, 113] using independent methods gave basically the same results regarding the electron excited states within the $V_k + e$ model, in disagreement with experiment.

The authors of Ref. 113 combined the CNDO and HADES methods in an effort to determine the adiabatic potential energy of the lattice containing a STE as a function of the Cl_2^- axial translation along the (110) axis. The HADES code was used to determine the lattice relaxation for each position of the V_k center. The ionic positions from the HADES calculation were then used to run the CNDO code to obtain the total energy of the cluster. In this way, they found that on the lowest energy STE surface (a_{1g} when centered), there was an energy barrier of about 2 eV to reach the nearest-neighbor F-H pair configuration. The ground state potential surface remained quite parallel to this excited state (in D_{2h} symmetry), and the recombination energy was found to be in good agreement with experiment. On the other hand, when the hole was excited to a b_u state, they found that the potential energy surface appeared flat, up to the nearest-neighbor (F-H) configuration.

As we will discuss in more detail later, this combining of the CNDO and HADES codes contains a shortcoming. When the lattice distortion was obtained with HADES, only the self-trapped hole (X_2^-) was present, but the excited electron was absent. An electron charge which is moderately diffuse, but interacting with a relatively small number of ions (about a few dozen) is not represented when the ionic displacements are determined. Therefore, the CNDO method as used may not describe the electron-lattice interaction properly.

In 1984, Song and co-workers presented new studies of the structure of the STE in alkali halides [115, 116] and in the fluorites [117]. They developed an efficient way of performing the extended-ion method of calculation. This was made feasible by the exclusive use of floating 1s Gaussians as the basis in describing the excited electron of a defect [118]. Various energy terms in the one-electron Hartree-Fock equation were replaced by accurate interpolation formulae which are made simple due to the use of 1s Gaussians. The lattice is represented by the traditional pair potentials and the electronic polarization is treated by a low order Mott-Littleton method. A relatively small region of the lattice surrounding the defect was allowed to relax explicitly (from about a dozen to several dozen ions). All ions inside the cluster are allowed to relax in turn so that the total energy of the system is minimized.

In the first work on NaCl and KCl [116], 12 ions were allowed to relax with complete freedom, while the electron was represented by about a half dozen floating 1s Gaussians distributed around the V_k core. When the system reached the minimum of the energy surface, the following was found: as soon as the minimization cycle starts, the electron wavefunction distorts and moves to one side of the V_k core, while at the same time the V_k core is expelled in the opposite direction. It was also found that the surrounding cations relax in such a way that they ease the passage of the V_k core on one side, while the cations at the side opposite to the former relax in such a way that they stabilize the emerging "F center electron". In this work, the equilibrium was reached after the V_k core had moved along the (110) direction by ≈ 3 a.u. in KCl and ≤ 3 a.u. in NaCl. An analogous calculation using the ion-size parameters *A* and *B* resulted in a similar structure [115]. Later, more work in other alkali halides confirmed that this off-center translational instability of the STE ($V_k + e$) system is general and quite strongly pronounced in all alkali halides [116, 119]. The off-center relaxation in KCl is illustrated in Fig. 24.

Work similar to this was also done on CaF_2 and SrF_2 [117], confirming theoretically that the STEs in those materials are indeed nearest-neighbor F-H pairs, which may assume two possible configurations in the lattice. This is in complete agreement with earlier experimental data [66, 120]. The V_k core, originally oriented along $\langle 100 \rangle$, is first expelled partially

along the $\langle 100 \rangle$ axis, and then rotation sets in to stabilize the F_2^- core approximately along the $\langle 111 \rangle$ axis. The electron takes over the vacancy and becomes an F center. It is this nearest-neighbor F-H pair configuration, illustrated in Fig. 25, which emits the intrinsic luminescence of self-trapped excitons in the fluorite crystals [120]. There is evidence that this instability is very general, and takes place in more complex materials such as SrFBr [121].

Before we analyze the various consequences of this off-center structure of the STE, it is necessary to understand how it happens. In Table 3 we have collected the calculated electron energies of F centers and of STEs (assumed $V_k + e$ structure) in several alkali halides from earlier publications [94, 95, 111]. These two defects were studied using identical methods. In all the cases, the electron energy is deeper in the F centers than in the STE ($V_k + e$) by about 3 eV. This indicates that the excited electron will seek to become an F center rather than staying in the system ($V_k + e$), if it has the choice. In a lattice, this can easily be achieved by a suitable lattice distortion which involves the expulsion of the V_k core away from the electron. The energy expended to accomplish this change is about 1.5 eV. Therefore the system gains about 1.5 eV net energy minimization and the off-center STE is realized. In these terms, theoretical evidence for the off-center structure of the STE has existed since the two works on relative energy levels of the F center [111] and of the on-center STE [94, 95] were published, quite apart from specific experiments or additional theoretical work.

It is always nice to be able to explain a result in more descriptive physical language. The origin of the instability of the ($V_k + e$) system can be made more transparent as follows: the reason a hole attracts an electron to form a STE in alkali halides is quite different from the case in rare-gas solids. The V_k core itself is repulsive to an excited electron in terms of its pseudopotential, as discussed in Ref. 116. It can be shown even more directly by an *ab initio* molecular calculation for ($X_2^- + \text{electron}$) [122]. The attractive potential is produced by the ionic lattice surrounding the V_k center. Therefore, the electron seeks to occupy a well-formed anion vacancy and expel the repulsive V_k core at the same time.

By contrast, in the case of excitons in rare-gas solids (one-center and two-center types), the ionized atom R^+ and the molecule R_2^+ are attractive for an electron. Although an evolution of ($V_k + e$) to (F-H) analogous to that in alkali halides is therefore not feasible, the instability that does occur is an interesting and instructive counterpoint. The light rare-gas solids such as neon and argon have negative electron affinity because the filled-shell rare-gas atoms represent net repulsive pseudopotentials for a conduction electron [46]. This is the origin of electron bubble formation. Energetics favor expulsion of the conduction electron from the crystal with about 1 eV of kinetic energy. If the excited ($\approx$ conduction) electron

is itself bound coulombically to an ionized rare-gas (R^+) at the surface, it may take the ion with it as it is expelled. This is essentially the mechanism proposed by Coletti *et al.* to explain the expulsion of excited rare-gas atoms from the surface of the light rare-gas solids [123]. On comparing this with the description of off-center STE relaxation in alkali and alkaline earth halides given above, one is struck by both its contrasts and similarities. The situations are almost mirror images. In the former case, the conduction electron is attracted to the lattice while being repelled by the core X_2^- . In the latter case the conduction electron is repelled by the lattice while being attracted by the R^+ or R_2^+ core. Instability appears to be the result in both cases.

It is also interesting to analyze the instability of the ($V_k + e$) system in alkali halides from the point of view of the pseudo Jahn-Teller effect. The electron states a_{1g} and b_{3u} of the ($V_k + e$) system interact with each other through a lattice vibrational mode of symmetry b_{3u} . The translational motion of the V_k core (called earlier the Q_2 mode [124]) and the odd symmetry mode of vibration of the nearest-neighbor cations (called $\bar{Q}_2$ mode) [125] are both of b_{3u} symmetry. For a strong Jahn-Teller effect between the two states, it is required that (i) the two states be close in energy, (ii) the coupling between the two states be

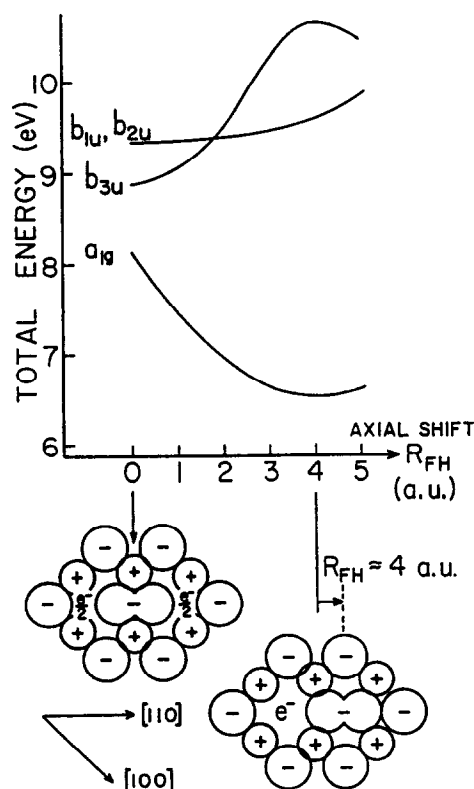


Fig. 24. Total energy (i.e. sum of the defect electron energy and the change of the lattice energy relative to the perfect lattice) vs the translational coordinate R_{FH} of the self-trapped exciton in states designated by the electron orbital in D_{2h} symmetry: a_{1g} , b_{3u} , b_{1u} , and b_{2u} (from Ref. 119).

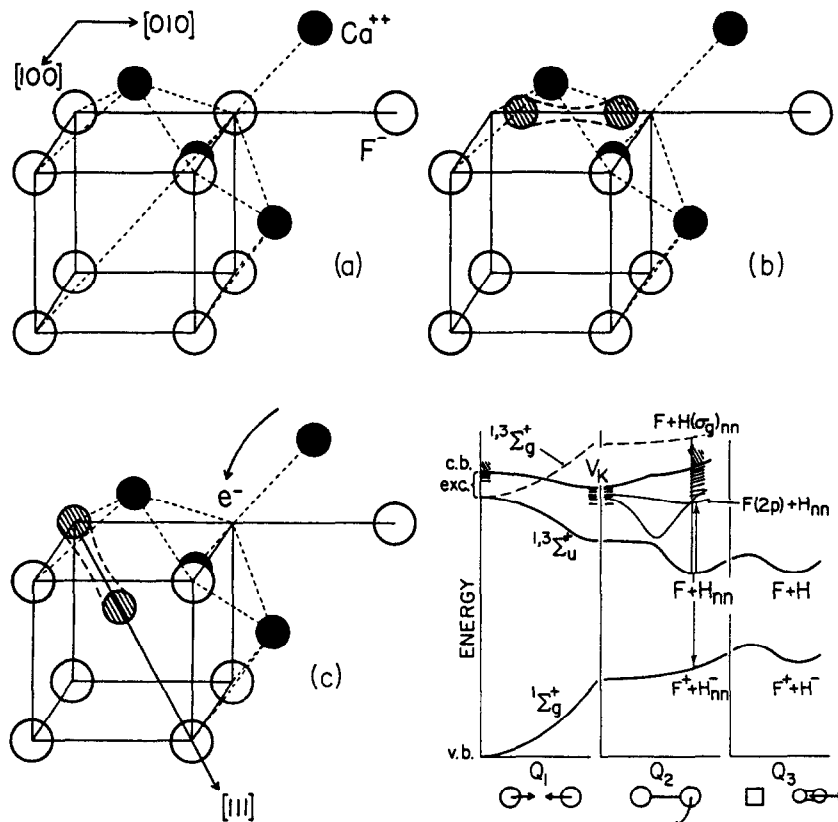


Fig. 25. Schematic illustration of exciton self-trapping in CaF_2 : (a) perfect lattice; (b) self-trapped hole; and (c) self-trapped exciton, showing how the (fluorine) $^-$ has rotated into an interstitial $[111]$ position to make room for the captured conduction electron to occupy the vacancy opened at the center of the calcium tetrahedron (from Ref. 103).

strong, and (iii) the Q_2 mode be soft. All three conditions are well satisfied in the case of the system ($V_k + e$) as was shown in earlier papers [124–126].

Toyozawa pointed out that the instability becomes strongest as the a_{1g} and b_{3u} states approach along an even-parity coordinate such as Q_1 , and suggested this as the origin of F center production from excited states of the STE [124]. The subsequent observations and calculations we have reviewed indicating spontaneous off-center relaxation in the lowest luminescent state of the STE imply that the b_{3u} – a_{1g} instability discussed by Toyozawa occurs more generally than

just near close approaches of the interacting energy surfaces. The a_{1g} surface is apparently unstable even at its minimum in D_{2h} symmetry [126].

The energies of the two interacting states a_{1g} and b_{3u} move in opposite directions as the off-center relaxation proceeds. The a_{1g} surface undergoes an instability because the energy of that state is depressed as off-center relaxation proceeds, while the b_{3u} energy level rises. This can explain why the b_{3u} sublevel of the p -like states is higher than the others according to experiment.

The off-center configuration of the STE requires the b_{3u} energy to be raised from its earlier calculated values, which were too low. In Table 4, we have collected the electron excited state energies obtained for ($V_k + e$) and for the STE at off-center equilibrium. Before, it was never possible to raise the calculated b_{3u} level sufficiently under the assumption that the STE occupies the ($V_k + e$) configuration.

Shluger and Kotomin [127] employed an INDO code to study the STE in KCl. They used a cluster containing 24 K^+ and 24 Cl^- ions, and the ionic positions were those published by Leung *et al.* [116]. Their calculation indicates that the vibrational mode representing the off-center motion of the V_k core in the STE is very soft, so that there is effectively no

Table 3. Comparison of calculated binding energies for F centers and for the ($V_k + e$) model of the self-trapped exciton†

	NaF	NaCl	KCl	KBr	KI
F	–6.9	–6.0	–6.0	–5.8	–5.4
$V_k + e$	–3.0	–2.8	–3.4	–3.2	–3.1
ΔE	3.9	3.2	2.6	2.6	2.3

† F-center binding energies are from Ref. 111. Calculations for the $V_k + e$ model of the self-trapped exciton are from Ref. 95 for sodium halides and from Ref. 94 for potassium halides. The differences in theoretical binding energies for the F center and ($V_k + e$) STE are listed as ΔE . Units are electron volts.

Table 4. Calculated and experimental absorption energies from the lowest electron orbital (a_{1g}) to the higher orbitals (b_{nu}), $n = 1-3^+$

	NaCl			KCl			RbCl		
	C_{2v}	D_{2h}	Exp.	C_{2v}	D_{2h}	Exp.	C_{2v}	D_{2h}	Exp.
$a_{1g} \rightarrow b_{1u}$	2.99	0.65	2.12	2.02	1.38	2.10	2.41	1.43	1.89
$a_{1g} \rightarrow b_{2u}$	2.77	1.42	1.95	2.29	1.94	1.87	2.24	2.49	1.71
$a_{1g} \rightarrow b_{3u}$	2.76	1.25	2.13	3.00	0.32	2.19	3.15	0.64	1.99

† Calculated values are given for both C_{2v} (off-center relaxed) and D_{2h} (on-center, $V_k + e$) symmetries [112]. Experimental energies are from Tanimura and Itoh [98]. Units are electron volts.

barrier to such relaxation. However, they do not find the strong potential gradient driving it off-center as found in the theory of Song *et al.* [115–119]. The difference seems due both to the inherent limitation of the INDO (also CNDO) methods and the size of the cluster used. In these semi-empirical methods, the valence electrons are represented only by the valence shells of the halide ions, while the excited electron is represented only by the alkali excited electron wavefunctions. As a consequence, the excited electron of the STE sees the alkali ions as naked point charges. If the effect of the cation cores were incorporated in the method, the excited electron wavefunction would be more diffuse and require a larger cluster. The on-center STE energy would rise higher leading to a stronger instability.

6. EXAMINATION OF OBSERVATIONAL DATA IN LIGHT OF OFF-CENTER AND $V_k + e$ MODELS OF SELF-TRAPPED EXCITONS

Whether the self-trapped exciton exists in an "off-center" or "on-center" configuration has obvious importance for any further work involving self-trapped excitons, certainly including defect formation by ionizing radiation. Although the calculations of Song *et al.* seem to clearly indicate off-center relaxation, this is a question that must finally be settled by experiment. Some of the experiments already discussed seem best explained by an off-center STE. There are also many experiments which have been quite well explained on the basis of the $V_k + e$ model.

The question of off-center relaxation of STEs in alkaline earth fluorides has been settled affirmatively by the definitive experiments on excited-state EPR [120]. The nuclear quadrupole resonance experiment of Block *et al.* indicates that the STE in KCl is not in an on-center $V_k + e$ configuration [101], but does not establish whether the degree of off-center relaxation is as large as is found in the theory of the alkali halides. Until a more definitive experiment on this question in the alkali halides is performed and widely accepted by workers in the field, the best appeal to experiment is an examination of the considerable body of data already accumulated, looking for consistency or inconsistency with one or the other of the models of the STE. Most of these comparisons have already been made at various times. This paper is an opportunity to bring them together in one place. The

review of observational data on the STE which we have already given in Section 4 makes it very convenient to present a systematic comparison of the data to the two theories here. We will proceed in the same order as the data were presented in Section 4, omitting the topics which have no bearing on the issue of off-centeredness.

(1) Luminescence spectra and polarization

In the $V_k + e$ model, the large Stokes shifts and widths of the STE luminescence bands are attributed to the steep ground state potential curve dominated by repulsion of the two filled-shell halide ions near the STE lattice configuration. This corresponds to the vibrational mode Q_1 discussed in Refs 124 and 126. In the off-center STE, the bound halide pair has essentially the same kind of Q_1 -mode repulsive ground state, so that part of the contribution to the Stokes shift and width is substantially the same in both models. Furthermore, the correlation between the calculated axial shift ΔQ_2 and Stokes shift [112] as a fraction of band gap reported in Table 5 suggests an additional contribution to the Stokes shift from the off-center relaxation mode Q_2 . To match the extreme Stokes shifts of the π luminescence, e.g. 9.4 eV or 70% of the band gap in LiF [55], the $V_k + e$ model invokes large Q_1 relaxation, as well as a large electron binding energy. Such a large binding energy for the on-center STE is not justified in view of the fact that the two-center STE has a more diffuse wavefunction than the one-center (free) exciton, as can be seen in the much smaller exchange splitting calculated [128]. Alternatively, in the off-center model one can appeal to both Q_1 and Q_2 relaxations to account for the Stokes shift and binding energy. Table 5 shows that the observed variation of Stokes shift among the alkali halides correlates with the predicted degree of off-center relaxation. Also included in Table 5 are entries for impurity-dimer associated self-trapped excitons ($I_2^- + e$ in KCl:I and RbCl:I) [129]. These systems behave like the self-trapped exciton in NaBr and NaI, characterized by smaller Stokes shift and shorter lifetime. It is apparent that in these systems also there is good correlation between the triplet state parameters and the off-center axial shift. The correlation between triplet STE lifetime (τ_π) and ΔQ_2 which is also shown in Table 5 will be discussed later.

Table 5. Experimental and theoretical parameters of triplet self-trapped excitons in alkali halides are compared with each other and with calculated values of the axial shift ΔQ_2 †

	π emission peak		Stokes shift (%)	Optical binding energy		Lifetime (10^{-6} s)	ΔQ_2 (a.u.)
	Exp.	Theory (eV)		Exp.	Theory (eV)		
NaF	2.5	1.9–2.7	76		2.37	6700	≥ 4
KF	2.48	2.04	75		1.60	38000	≥ 4
RbF	—	1.64	—		1.98	—	≥ 4
NaCl	3.47	3.36	57	2.8	3.28	295	≤ 2
KCl	2.54	2.7–3.4	68	2.6	2.42	5000	≥ 5
RbCl	2.41	2.3–3.1	69	2.3	2.24	5500	≥ 5
NaBr	4.65	3.21	31	≤ 1.1	3.04	0.49	≤ 2
KBr	2.44	3.09	65	2.3	2.35	130	≥ 4
RbBr	2.36	2.56	66	2.2	2.44	180	≥ 4
NaI	4.22	2.98	24	≤ 1.2	4.32	0.09	≤ 2
KI	3.36	3.68	42	1.7	2.07	4.4	≤ 2
RbI	3.09	3.56	47	1.7	1.98	11	≤ 3
(I $_2^- + e$)							
KCl:I	4.64	4.64	28			0.105	≈ 1
RbCl:I	4.39	4.01	35			0.122	≈ 1

† The optical binding energy is deduced experimentally as the limit of the Rydberg-like series of electron transitions in absorption [61]. The Stokes shift of π luminescence is expressed as $\{E(1s) - E(\pi)\}/E(1s)$, where $E(1s)$ is the energy to create a free exciton in the pure crystal, or to create an I^- excitation in the mixed crystals; and $E(\pi)$ is the peak energy of the STE π luminescence. Experimental data for chlorides, bromides, and iodides are from Refs 50 and 61; for fluorides, from Ref. 55; and for (I $_2^- + e$) in chloride hosts, from Refs 72 and 129. Theoretical values were obtained for the off-center model by Song and Leung in Ref. 112.

Suzuki *et al.* [130] observed that there is a low energy tail which is part of the STE π emission band in NaCl. This tail grows as the temperature is raised, and is accompanied by a blue shift of the emission band. The authors conclude that the most reasonable way to understand this result is to assume second-order coupling of the exciton luminescent state with a very soft mode ($h\nu = 2.2$ meV). By identifying the soft mode with the axial translation of the V_k core, or Q_2 mode, they deduced that the STE is shifted off-center by about $1 \text{ \AA}$ at equilibrium in the π luminescent state.

Polarization of the STE luminescence relative to the V_k molecular bond axis was described very successfully within the $V_k + e$ model by Kabler [48], Murray and Keller [49], and in the theory of Fowler *et al.* [131]. Detailed treatments of polarization of luminescence from the off-center STE have not been given. In general, the polarization properties should be similar for the two models in alkali halides if the off-center motion of the V_k core is along the (110) direction of the V_k bond axis and if the orientation is maintained along that axis. In that case, the F-center-like electron charge distribution is still symmetrically distributed about the X_2^- bond axis direction, so the polarization properties characterized as sigma and π relative to the V_k bond axis and determined essentially by the hole state, will not be significantly different in the off-center and on-center pictures. The electron-hole overlap is certainly different, but this mainly affects the lifetime, to be discussed later.

In many alkali halides, no σ -polarized emission band with a short lifetime is observed. Earlier, it was argued that this was likely a problem of kinetics

of population of the singlet STE [132]. From preliminary work [133], it seems that the higher excited states are on-center, in particular the b_{3u} and a_{1g}^* states. Then the previous assignment of a_{1g}^* for the initial electron orbital for the σ luminescence transition would place the emission energy too high. One speculation is that there may be a local energy minimum on the b_{3u} state along the Q_2 axis, off $Q_2 = 0$. If so, the dipole transition would become allowed to the ground state by the same reason as it is allowed from the off-center a_{1g} state. At the same time the transition energy would be in the right range, about 4 eV. Such an idea is being pursued at present.

There is the possibility that at temperatures above the H-center reorientational threshold, the X_2^- core of the off-center STE may be able to rotate out of the line connecting the "F center" and "H center" parts of the off-center STE. It has been speculated that this may account for a temperature-dependent change in the degree of π -polarization of STE luminescence in KCl [106]. Alternatively, the temperature-dependent π -luminescence polarization in KCl has been attributed to thermal excitation from the triplet state to a singlet lying about 15 meV higher [132].

(2) Quenching of luminescence

It was shown in Refs 75–78 that quenching of STE luminescence must involve activation over a barrier against F–H pair formation in crystals such as KI, NaI, NaBr, and NaCl which exhibit correlation between luminescence quenching and F center formation. The activation process occurs on the adiabatic potential surface that encompasses both

STE and F-H pairs, rather than crossing to the ground state as previously assumed. It was shown in Ref. 119 that this process would be facilitated by a tendency of the STE to move off center, in order to account for the unexpectedly small barrier height. Without the assistance provided by the electron binding energy at the F-center site as discussed earlier, the barrier to STE/F-H conversion posed mainly by the central cation pair is found to be prohibitive [113].

(3) Lifetime

The approximate proportionality of STE π luminescence decay rate and halogen spin-orbit coupling was well accounted for by Kabler and Patterson using the $V_k + e$ model [50]. This dependence on the halogen spin-orbit coupling and its consequent admixture of allowed ${}^1\Pi_u$ character into the forbidden ${}^3\Sigma_u^+$ lowest excited state holds for the off-center STE as well. In addition, however, the lifetime of the off-center STE should scale inversely with the amount of Q_2 relaxation, which tends to separate the electron and hole wavefunctions. In Table 5, the π luminescence lifetime, Stokes shift, and calculated Q_2 relaxation are compared. There is good correlation between the lifetime and the degree of Q_2 relaxation calculated in the off-center model, as the alkali metal changes within each halide group. That is, for a given halide, heavier alkali corresponds to larger (calculated) off-center relaxation of the STE, larger Stokes shift, and longer lifetime.

(4) Excited state absorption and double excitation spectroscopy

These measurements together provided the energies, ordering, and polarization of transitions to the higher "excited electron" STE states from the lowest luminescent level. An important test of any model of the STE is its ability to predict these levels correctly. Calculations in which the STE stays on center (D_{2h} symmetry) have consistently predicted that the ordering of excited electron orbitals should have b_{3u} lower than b_{1u} and b_{2u} , and have found a transition energy to b_{3u} consistently 2 or 3 times lower than the lowest observed transition. One of the clearest successes of the off-center STE model is that it gets the ordering and approximate energies of the excited electron states right. As the halogen core moves off center, breaking the D_{2h} symmetry, the lowest electron orbital (a_{1g} in D_{2h} symmetry) drops significantly in energy as the electron moves to occupy a mixed state of the developing vacancy. The lowered symmetry brings mixing of the a_{1g} and b_{3u} orbitals, with the b_{3u} rising above the b_{1u} and b_{2u} electron orbitals. In a simple F center, all three b_u states would be degenerate. An atomistic reason for why b_{3u} turns out to be the highest of the b_u states in the off-center STE is that the b_{3u} electron orbital sees the repulsive X_2^- along an orbital lobe, while the other two see the X_2^- on the nodal plane of their wavefunctions.

From the excited-state absorption spectra, the ionization limit of the Rydberg-like sequence of excited-electron states can be identified. The observed ionization limits for triplet self-trapped excitons are typically 2 or 3 times larger than the ionization limit of the corresponding free exciton [61]. The observed STE ionization limit is closer to that of the F center than of the free exciton. This is the prediction of the off-center STE model, in which the electron occupies a partial vacancy approximating an F center. Table 5 compares the experimental STE ionization limits with the theoretical predictions of the off-center model.

(5) Optically-detected EPR

As noted earlier, optically-detected EPR showed that the two halogen nuclei comprising the core of the STE have nearly equal hole densities, as evidenced by the V_k -like (or H-like) hyperfine pattern. This observation is obviously consistent with the on-center STE, or $V_k + e$ model. It can also be consistent with a nearest-neighbor F-H pair in which the F-center electron is localized on the vacancy at one end of the H center. The trapped electron restores one negative charge to the vacancy so that each anion site surrounding the H center has an equivalent charge. If the halogen displacement were only part-way to the $(F-H)_{nn}$ configuration, the electron could not fully occupy the developing F center and the self-trapped hole would then be subject to distinct lack of mirror symmetry in the charge distribution, in contradiction to the EPR results. The existence of the EPR data requires that the off-center STE relaxation be either very large (about 4 a.u., close to the value for an F-H pair) or very small.

We can be somewhat more quantitative about this. As discussed in Ref. 119, the sum of the distorted-lattice Madelung potential and the potential due to the trapped electron differs by no more than about 0.1 eV at the two central halogen sites in the off-center STE, when evaluated for the full 4-a.u. equilibrium displacement calculated in KCl. Corresponding to this potential difference between the two sites, we find hole occupation probabilities of 48 and 52%, respectively, on the two halogen sites. This would split the central peak of the Cl_2^- hyperfine pattern into four subpeaks spanning about 24% of the separation between adjacent main-peak envelopes. Likewise, the adjacent main peaks would be split into three subpeaks spanning 18% of the main-peak separation, etc.; but in Fig. 23 showing optical EPR data for the STE in KCl, each of the seven hyperfine peaks has a width of 40–50% of the separation between adjacent peaks. Hence, the expected splitting of hyperfine components would not be resolvable against the broadening which has already occurred due to the interactions of the F-center-like electron with surrounding nuclei. Support for this hypothesis is given by failure to observe hyperfine splitting due to the naturally occurring

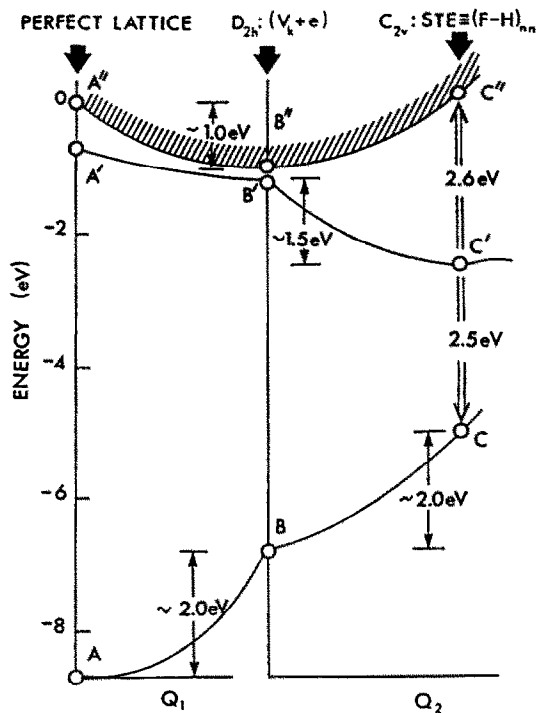


Fig. 26. Configuration coordinate diagram for the ground state and lowest triplet excited state of the self-trapped exciton in KCl, including relaxation from D_{2h} to C_{2v} symmetry (from Song and Leung, Ref. 112).

isotopes ^{35}Cl (75% abundant) and ^{37}Cl (25% abundant). These isotopes differ in nuclear moment, and hence in hyperfine constant, by 20%. By inference, then, a hole imbalance as large as 20% of the mean occupation on the two chlorine nuclei might not be detected in the hyperfine pattern.

The calculated Q_2 axial shift in the alkali iodides and in NaBr is smaller than in other alkali halides (Table 5). An asymmetry of the halogen sites should be observable in these materials if the self-trapped exciton moves only moderately off-center. Optically-detected EPR of the self-trapped exciton has, however, not yet been measured in these materials.

(6) Optically-detected nuclear quadrupole resonance

The observed four-line quadrupole splitting pattern of the STE in KCl implies two different quadrupole splittings for the central chlorine nuclei, and hence inequivalent sites. The $V_k + e$ or on-center STE is simply inconsistent with this observation. The off-center STE as described by Song *et al.* agrees qualitatively, i.e. in providing the inequivalent nuclei. Song and Leung calculated the electric field gradient on the STE core for their calculated off-center configuration of the STE in KCl and found satisfactory agreement with the experimental values [125]. Optical ENDOR of the self-trapped exciton in other alkali halides would provide important new data on this question.

(7) Summary

In summary, the experimental results on quadrupole splitting and some aspects of the excited state absorption data, particularly ordering of the b_u sub-levels, appear inconsistent with the on-center $V_k + e$ model of the STE. In addition, some recent experiments in spectroscopy, such as the π luminescence band shape in NaCl [130] seem to favor an explanation involving an off-center STE. Lifetimes of STE luminescence are not inconsistent with either model, although the off-center STE provides a better accounting for lifetime variation as a function of alkali constituent, in terms of the electron-hole overlap determined by the degree of off-center relaxation. Optically-detected EPR of the STE presents mixed results. The isotropic hyperfine splitting found in optical EPR and ENDOR agrees very well with the F-center-like nature of the electron orbital in the off-center STE. On the other hand, the hyperfine pattern indicates equivalent halogen nuclei within the resolution limits imposed by the hyperfine linewidth. Equivalency of the halogen nuclei corresponds to the on-center model. Near equivalency results if the off-center relaxation goes as far as the F-H pair.

By a tally of about 3 to 1, we believe the data discussed above are more consistent with an off-center self-trapped exciton. More definitive experiments are needed.

Designations of electron and hole orbitals and states of the self-trapped exciton in C_{2v} symmetry were summarized in Fig. 4, presented earlier, showing parentage in $D_{\infty h}$ and D_{2h} symmetries. We show in Fig. 26 a configuration coordinate diagram appropriate to C_{2v} symmetry of the self-trapped exciton in KCl. Note the similarity of some aspects to the configuration coordinate diagram for relaxation to C_{1h} symmetry of the self-trapped exciton in CaF_2 , shown in Fig. 25.

7. DEFECT FORMATION

It is well-established that vacancy-interstitial pairs (formed initially as F-center/H-center pairs) evolve from self-trapped excitons in at least the alkali halides and alkaline earth halides. The defect formation process is the subject of the following paper by Itoh and Tanimura.

Some of the most prominent features in the F-H pair formation process can be understood in a consistent and simple way on the basis of the off-center structure of the STE. One of the most remarkable results of Ref. 78 is that the STE and F-H pair are separated by a very small energy barrier (roughly 100 meV), while the $V_k + e$ model expects this barrier to be about 2 eV [113]. The small barrier is not at all surprising in the context of the off-center STE, but is otherwise difficult to understand. Calculation now shows that indeed the barrier is very small [134]. Also, the well-known Rabin-Klick criterion for F-center yields at low temperature [135, 136] can be

interpreted in terms of the off-center axial shift ΔQ_2 and the available relaxation energy of off-center STE motion along the (110) axis [134]. A similar correspondence can be drawn to the occurrence of energetic halogen atom emission from the surface of alkali halides in which the energy of Q_2 relaxation is large [137]. Further discussion of the relation of the off-center model of self-trapped excitons to defect formation has been given in Ref. 119.

Note added in proof—K. Kan'no and co-workers [138] have presented evidence that σ and π emission bands coincide in NaI and NaBr, and furthermore have proposed a classification of STE σ and π luminescence bands in three families among the alkali halides. In this scheme, the σ bands generally, and the corresponding π bands in NaI and NaBr, are suggested to originate from an on-center configuration. The low energy π bands generally, and the E_x bands in alkali iodides, are suggested to originate from two separate off-center configurations, all on the same adiabatic potential surface. Y. Kayanuma has also discussed such a classification [139]. This proposal brings a pleasing orderliness to the occurrence and peak energies of σ and π STE luminescence in alkali halides, and represents a new direction of investigation.

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